

ECE 105: Introduction to Electrical Engineering

Lecture 14

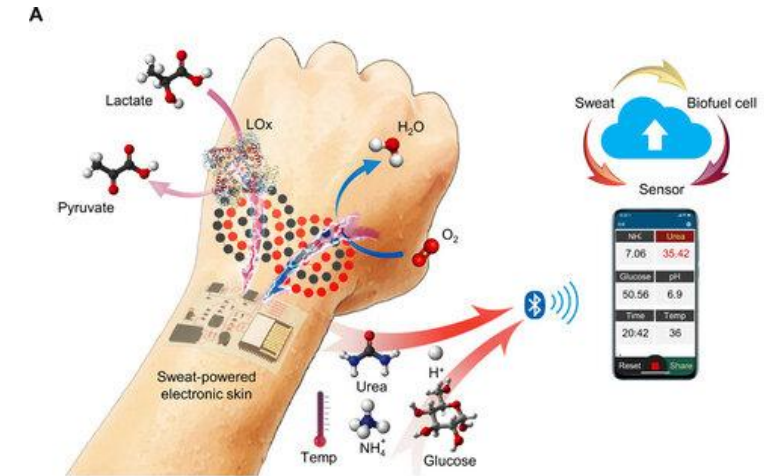
Sensors 2

Yasser Khan

Rehan Kapadia

Wearables

- Wearables include devices for communication, health, fitness tracking, etc.
- Future generations of devices will have new capabilities (therapeutic, haptic, etc.)
- New functionality depends on energy harvesting, conformal contact, etc.
- Wireless communication essential for most important functions (interfacing to mobile devices, etc.)



Vital signs

- Vital signs include Heart Rate, Blood Pressure, Breathing Rate, Temperature
- Attractive for daily monitoring
- Combining all of these metrics is more useful for preventive care



\$380.5 billion by 2028 – wearables alone

Global Wearable Medical Devices Market 2020-2024

Market growth will **ACCELERATE**
at a **CAGR** of almost

10%



Incremental growth (\$B)

\$10.24

Growth for 2020



13.88%



35%

Of growth will come from
NORTH AMERICA



The market is **CONCENTRATED**
with several players occupying the
market share

IMPACT



Health Care Industry:

The Health Care sector will see **POSITIVE IMPACT** due to the COVID-19 outbreak, and the industry is expected to register a high growth rate compared to the global GDP growth.



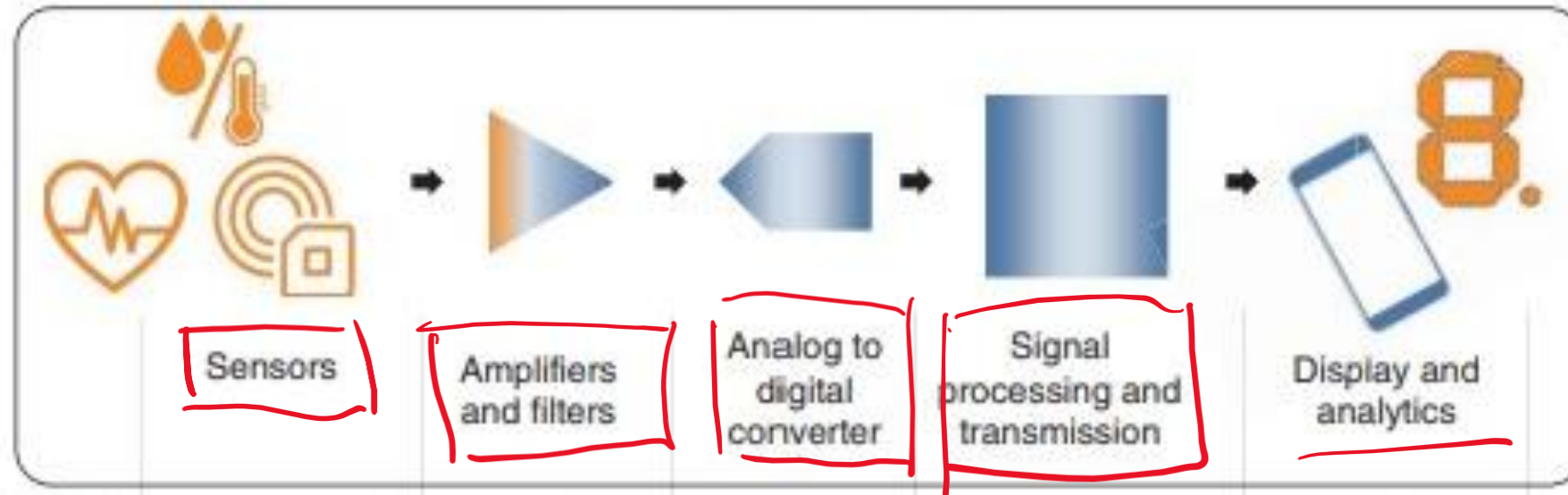
Market impact:

This market will have **POSITIVE IMPACT** due to pandemic



All market estimates to be revisited and updated in Q4-2020, based on the revaluation of the impact, as the pandemic spread plateaus

Different parts of a bioelectronics system



Pulse oximetry



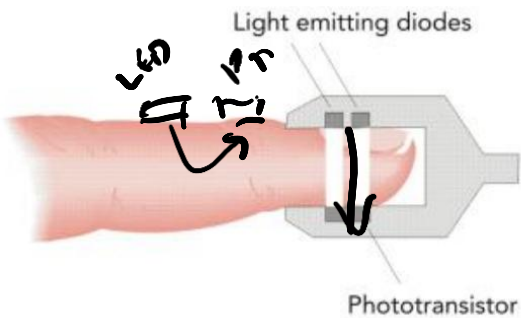
ECG

Pulse Ox

Respiration

Blood Pressure

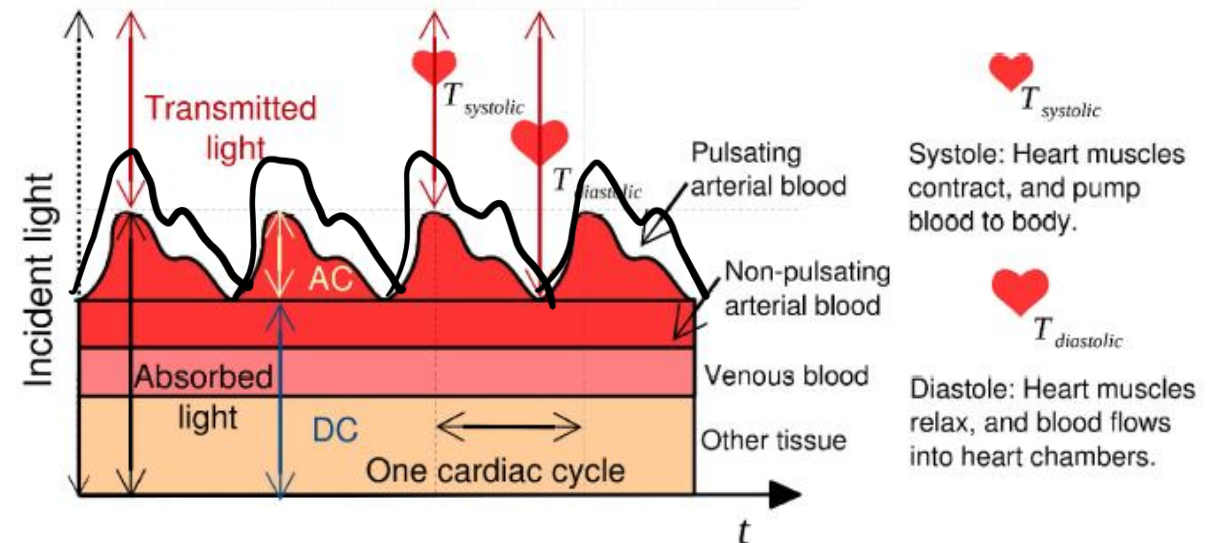
Temperature



$$SO_2 = \frac{C_{HbO_2}}{C_{HbO_2} + C_{Hb}}$$

- SO_2 The saturation of oxygen in blood,
- C_{HbO_2} Concentration of oxygenated hemoglobin (HbO_2),
- C_{Hb} Concentration of deoxygenated hemoglobin (Hb).

- Pulse oximetry measures blood oxygenation. Using spectrophotometry of absorptivity of blood at two distinct wavelengths, blood oxygen saturation is quantified.
- Can detect hypoxemia, ie. lower than normal blood oxygenation.



Pulse oximetry basics

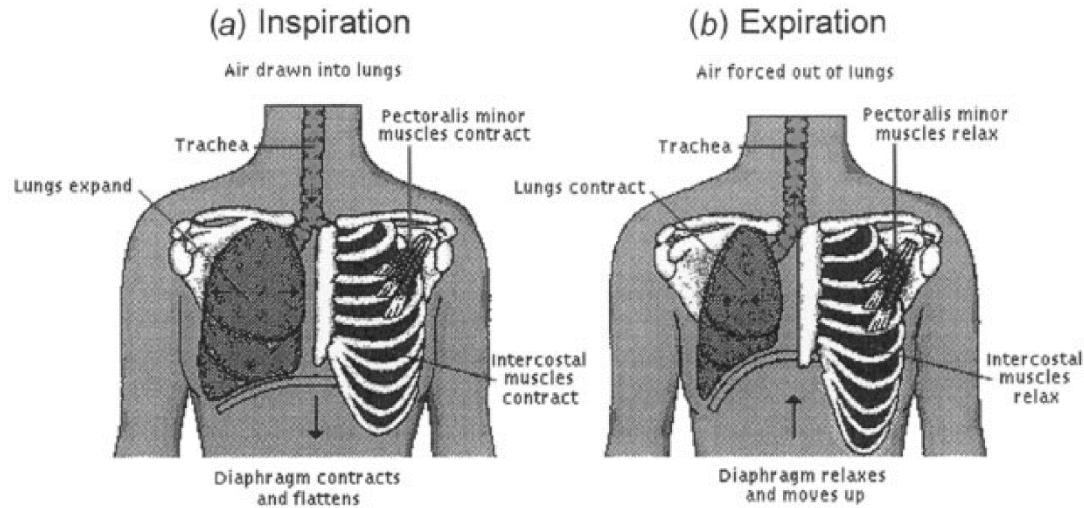


Figure 1.1 During inspiration, (a), the diaphragm, intercostal muscles and pectoralis minor muscles contract, causing the lungs to expand and air to enter the lungs. As the diaphragm, intercostal muscles and pectoralis minor relax, the lungs contract, causing air to leave the lungs (b), which is referred to as expiration (from Microsoft Encarta).

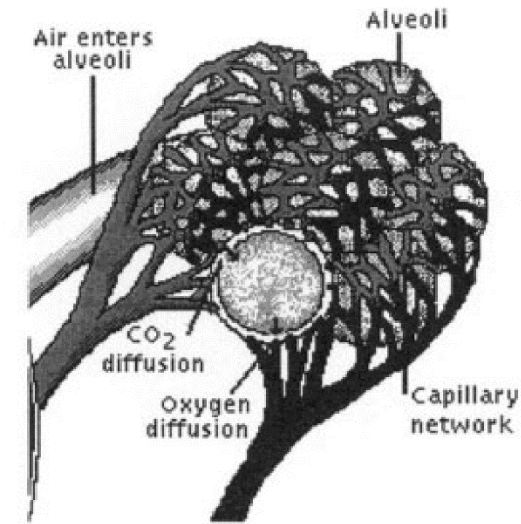


Figure 1.4 The capillaries surround the alveoli, providing the close proximity necessary for diffusion. Carbon dioxide diffuses from the capillary into the alveoli and oxygen diffuses into the blood (from Microsoft Encarta).

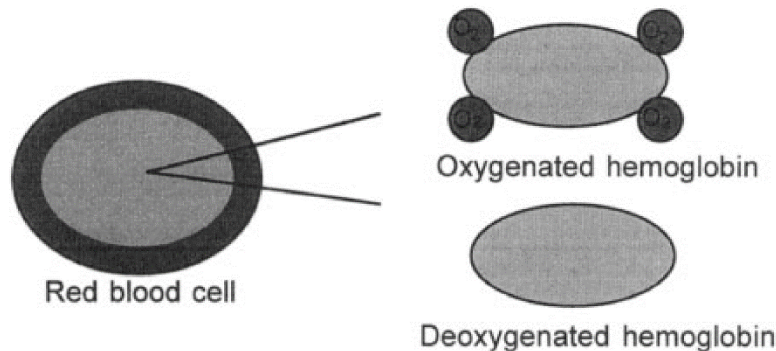


Figure 1.5 Hemoglobin molecules are contained within red blood cells. Each red blood cell contains approximately 265 million molecules of hemoglobin.

Pulse oximeter basics

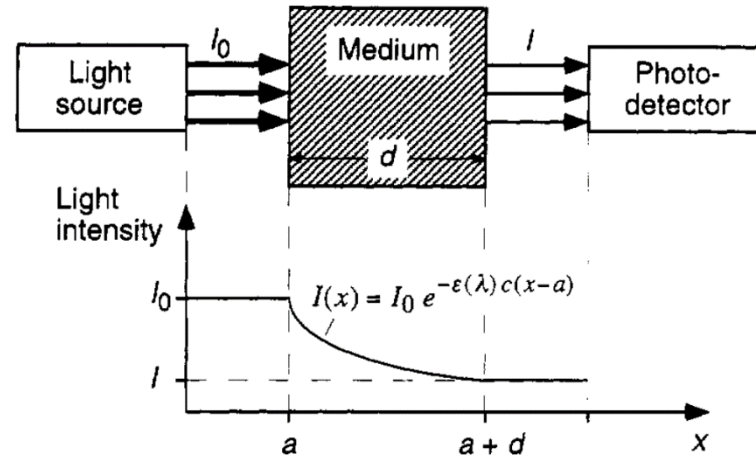
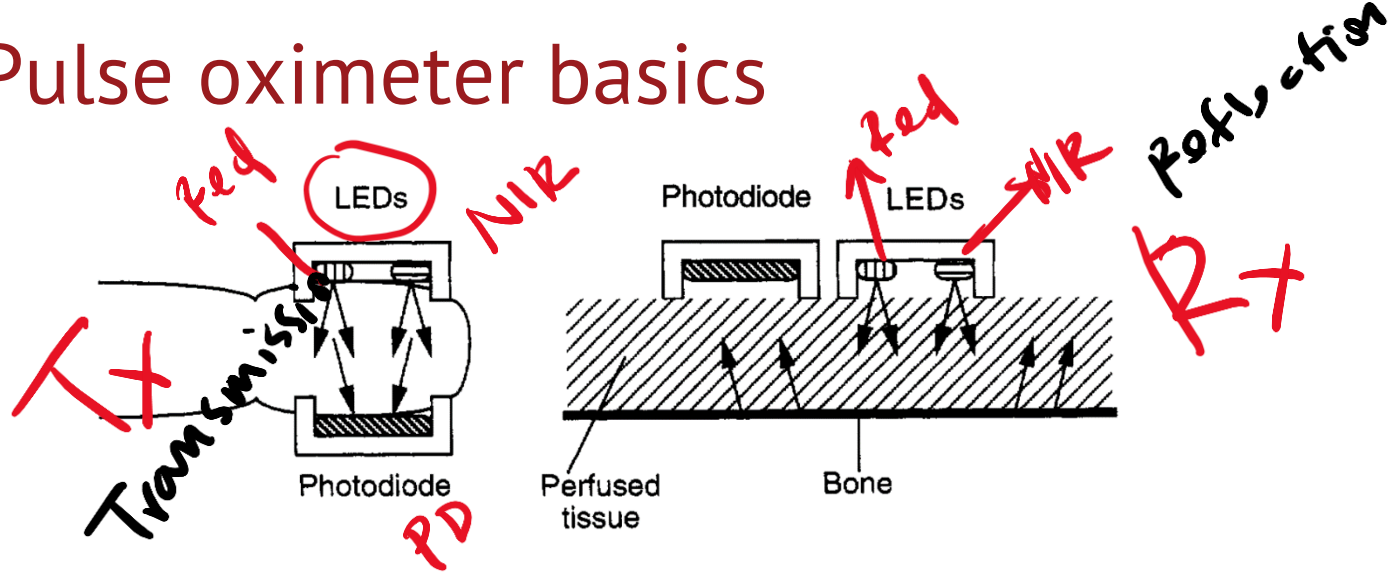


Figure 4.1 Beer's law: Incident light of intensity I_0 travels the distance a from a light source to the medium without being absorbed in the air. The light intensity decreases exponentially with distance in the absorbing medium. The intensity of the transmitted light I is determined by Beer's law. It stays constant after exiting the medium with optical path length d and can be measured by a photodetector.

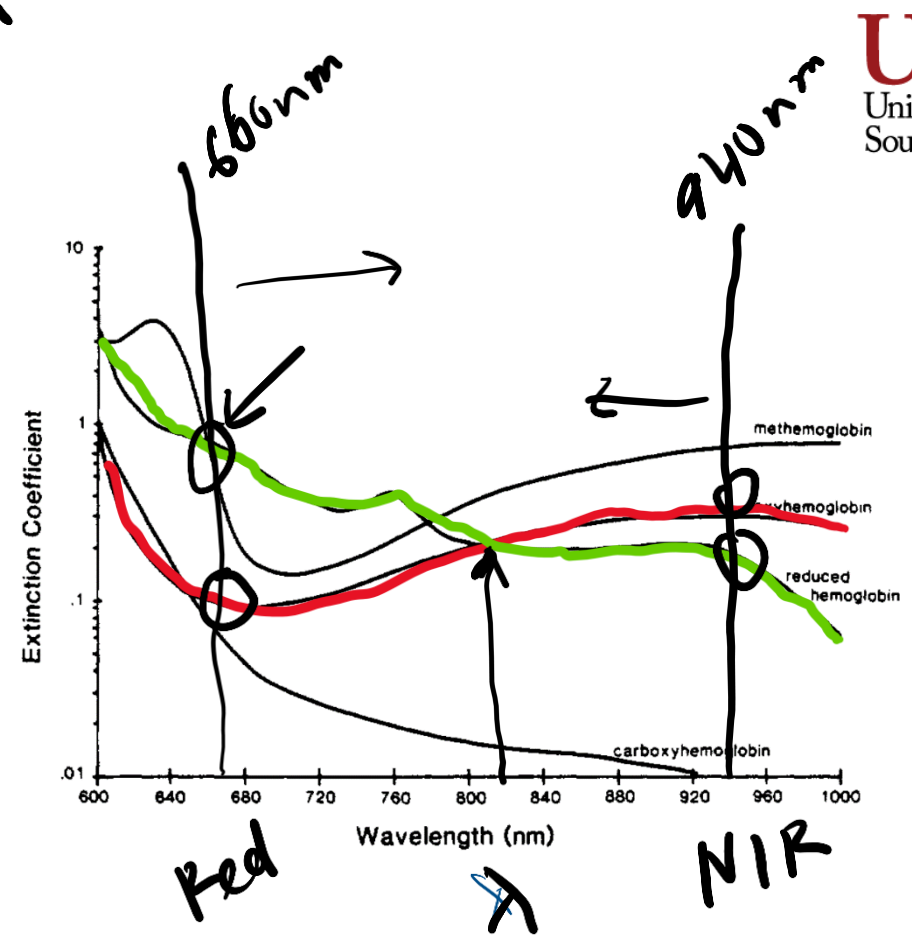
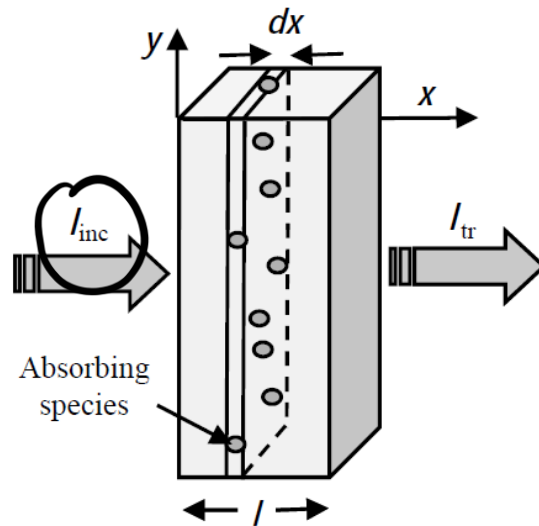


Figure 4.2 Extinction coefficients of the four most common hemoglobin species oxyhemoglobin, reduced hemoglobin, carboxyhemoglobin, and methemoglobin at the wavelengths of interest in pulse oximetry (courtesy of Susan Manson, Biox/Ohmeda, Boulder, CO).

The Beer-Lambert Law



A sample of length l , containing resonating absorbing species, subjected to incident light of intensity I_{inc} , exits the sample as transmitted light of intensity I_{tr} . An infinitesimally thin slice dx absorbs light of intensity dI .

Beer-Lambert (decadic)

$$I = I_0 10^{-\varepsilon c \ell}$$

- Transmittance: $T \equiv \frac{I}{I_0} = 10^{-\varepsilon c \ell}$
(Percent: $\%T = 100 T$)
- Absorbance (decadic): $A \equiv -\log_{10} T = \varepsilon c \ell$
(Also $A = 2 - \log_{10}(\%T)$ when $\%T$ is in percent.)

Definitions

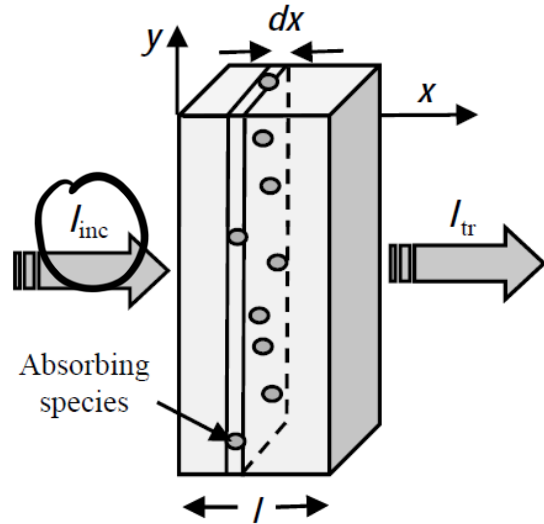
- Absorption (process): the medium removes energy from the light beam at a given wavelength.
- Absorbance (A): dimensionless log measure of attenuation, proportional to c and ℓ .
- Transmittance (T): fraction of light that passes, 0–1.

Units

- ε : molar absorptivity, $\text{M}^{-1} \cdot \text{cm}^{-1}$ (or $\text{L} \cdot \text{mol}^{-1} \cdot \text{cm}^{-1}$)
- c : concentration, M ($\text{mol} \cdot \text{L}^{-1}$)
- ℓ : path length, cm

Mixtures obey additivity: $A_{\text{total}} = \sum_i \varepsilon_i c_i \ell$ (same ℓ).

The Beer-Lambert Law



A sample of length l , containing resonating absorbing species, subjected to incident light of intensity I_{inc} , exits the sample as transmitted light of intensity I_{tr} . An infinitesimally thin slice dx absorbs light of intensity dI .

$$I_{inc} \rightarrow \boxed{} \rightarrow I_{tr}$$

ext. coeff. (conc.) length

$$- \epsilon [c] l$$

$$I_{tr} = I_{inc} 10$$

Transmittance, $T = \frac{I_{tr}}{I_{inc}}$

Absorbance, $A = -\log_{10}(T)$

Example 1

The intensity of light of wavelength 254 nm is attenuated to 16% of its incident value after passing through an alcohol solution of 50 mM benzene contained in a 1.0 mm thin quartz cuvette. Calculate the absorbance A and the molar absorption coefficient ϵ . What would be the transmittance through a 2.0 mm thick cuvette?

$$\lambda = 254 \text{ nm}$$

$$I_{tr} = 0.16 I_{inc}$$

$$[c] = 50 \text{ mM}$$

$$l = 1 \text{ mm}$$

$$A = ?$$

$$\epsilon = ?$$

$$T_{2\text{mm}} = ?$$

$$T = \frac{I_{tr}}{I_{inc}} = 0.16$$

Example 1

The intensity of light of wavelength 254 nm is attenuated to 16% of its incident value after passing through an alcohol solution of 50 mM benzene contained in a 1.0 mm thin quartz cuvette. Calculate the absorbance A and the molar absorption coefficient ϵ . What would be the transmittance through a 2.0 mm thick cuvette?

$$\begin{aligned} \lambda &= 254 \text{ nm} & I_{\text{tr}} &= 0.16 I_{\text{inc}} & [c] &= 50 \text{ mM} \\ l &= 1 \text{ mm} & A &= ? & & \\ & & \epsilon &= ? & & \\ & & A &= \epsilon \cdot [c] \cdot l & & \\ & & \epsilon &= \frac{A}{[c] \cdot l} & & \\ & & & \frac{0.8}{50 \text{ mM} \cdot 1 \text{ mm}} = 160 \text{ cm}^{-1} \text{ M}^{-1} & & \\ & & T_{2 \text{ mm}} &\Rightarrow & & \\ & & & A = 0.8 \times 2 = 1.6 & & \\ & & & 1.6 = -\log_{10}(\dots) & & \end{aligned}$$

Example 1

The intensity of light of wavelength 254 nm is attenuated to 16% of its incident value after passing through an alcohol solution of 50 mM benzene contained in a 1.0 mm thin quartz cuvette. Calculate the absorbance A and the molar absorption coefficient ϵ . What would be the transmittance through a 2.0 mm thick cuvette?

$$\lambda = 254 \text{ nm} ; \frac{I_{tr}}{I_{in}} = 0.16 = T ; [c] = 50 \text{ mM} ; l = 1 \text{ mm}$$

$$\boxed{A = ? \quad \epsilon = ? \quad T_{2 \text{ mm}} = ?}$$

$$A = -\log_{10}(T) = -\log_{10}(0.16) = 0.8$$

$$A = \epsilon [c] l \Rightarrow \epsilon = \frac{0.8}{50 \text{ mM} \cdot 1 \text{ mm}} = 160 \text{ cm}^2 \text{ M}^{-1}$$

$$\begin{aligned} T_{2 \text{ mm}} \Rightarrow A &= \epsilon [c] \times 2 \text{ mm} \\ &= 0.8 \times 2 = 1.6 \\ 1.6 &= -\log_{10}(T) \end{aligned}$$

basics

fed

feed ac

feed dc

feed ac

feed dc

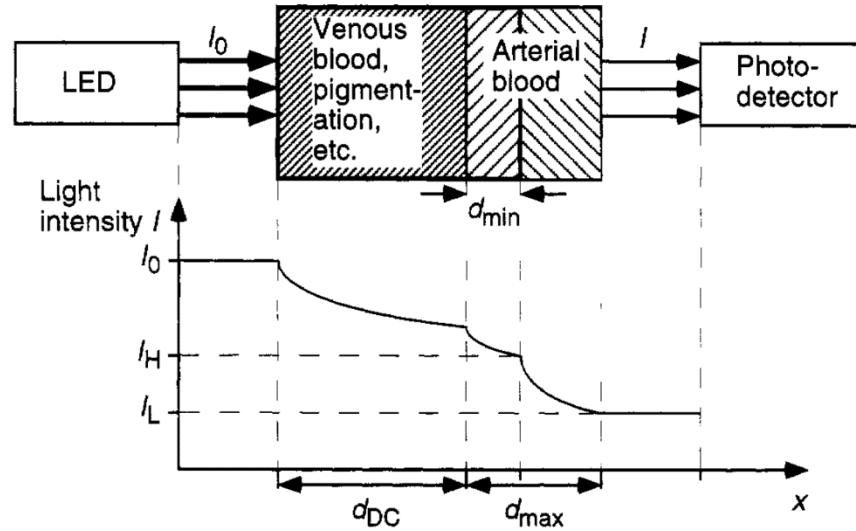
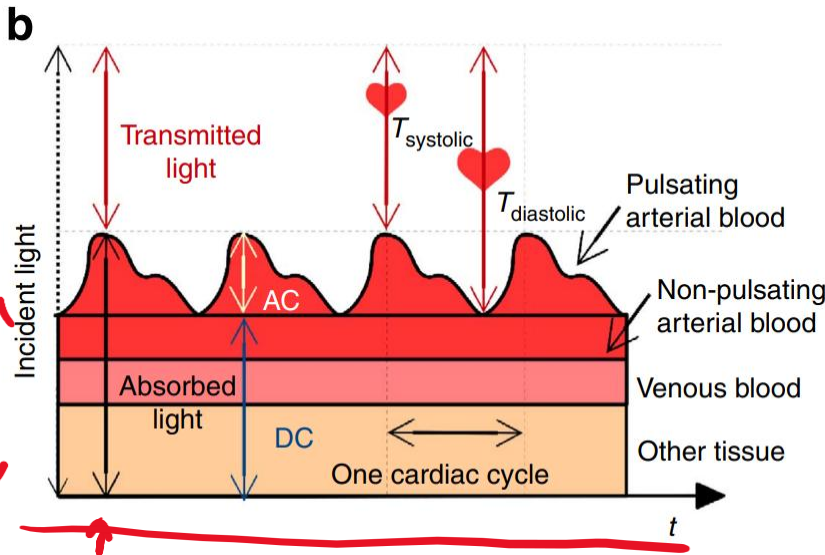


Figure 4.5 Beer's law in pulse oximetry. The DC components of the tissue (e.g. skin pigmentation, bone, venous blood and the nonpulsating part of the arterial blood) absorb a constant amount of the incident light I_0 . The effective optical path length in the DC components without the constant level of arterial blood is represented by d_{DC} . During diastole the optical path length through the arteries has a minimum length of d_{min} , and the light intensity at the photodetector is maximal (I_H). The optical path length reaches a maximum d_{max} during systole and the hemoglobin in the arteries absorbs a maximum amount, causing I to decrease to a minimum level of I_L .

Pulse oximeter basics

$$SpO_2 = f(R_{os})$$

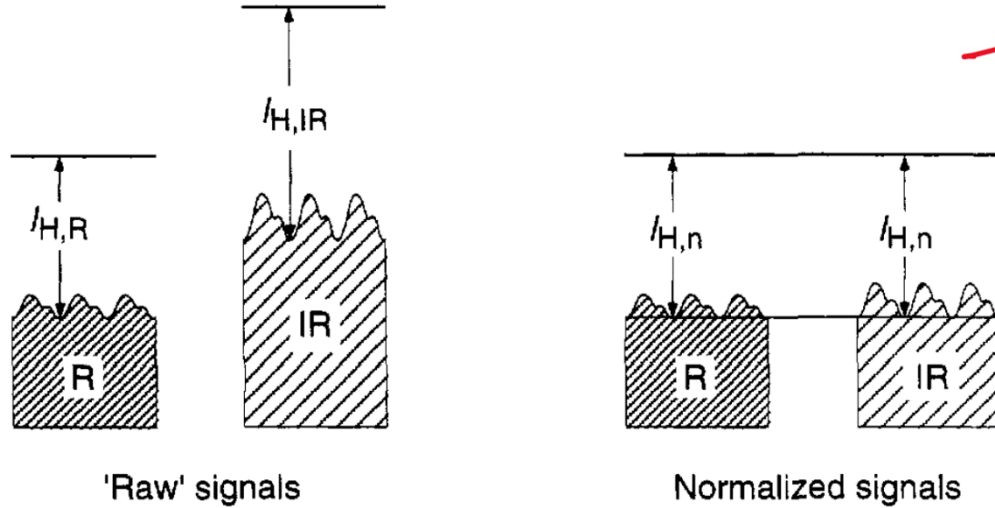


Figure 4.6 The normalization of the signals. The transmitted light from the red LED (R) and from the infrared LED (IR) is divided by its individual DC component. Thus, both normalized light intensities have the same magnitude during diastole. The normalized signals determine the basis for the calculation of the arterial oxygen saturation.

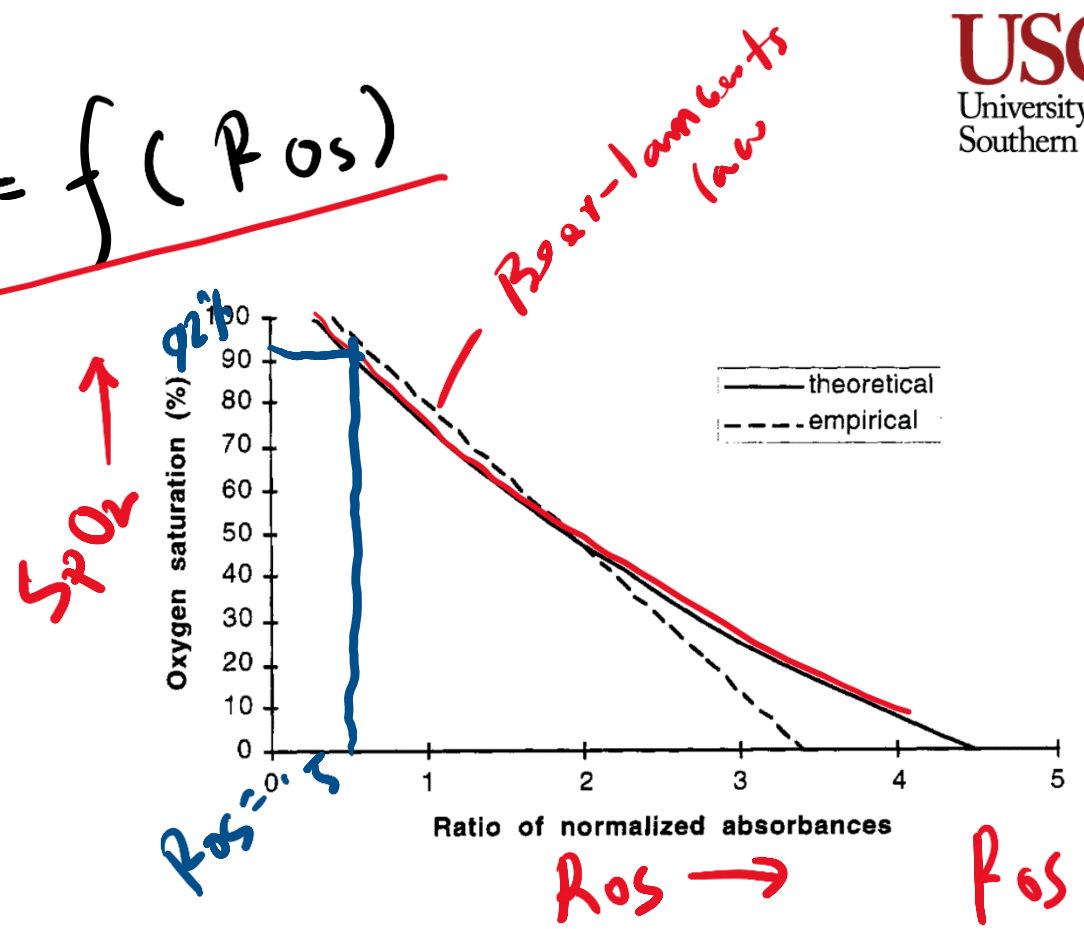
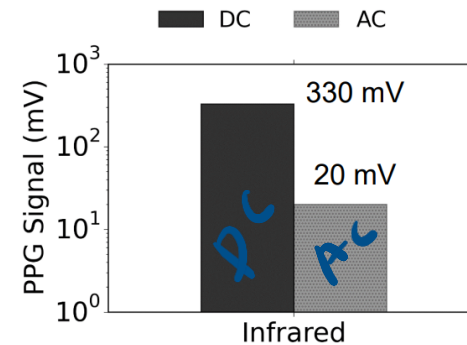
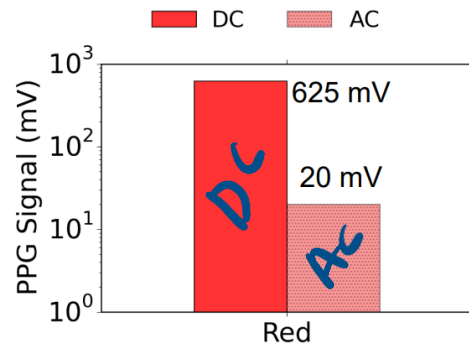
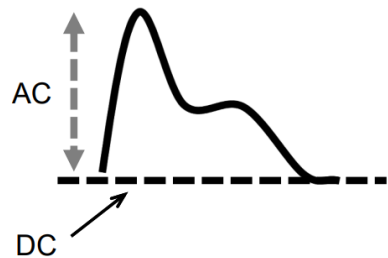
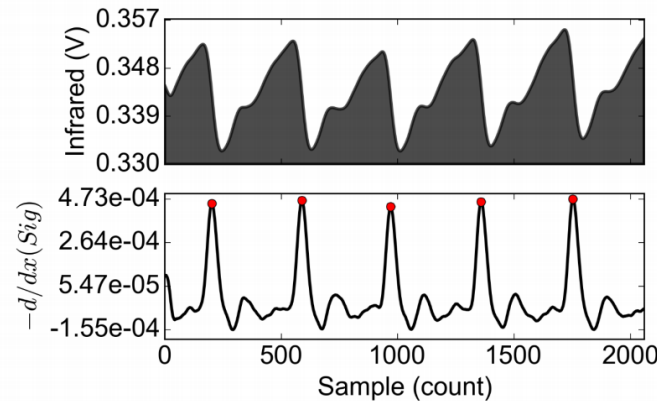
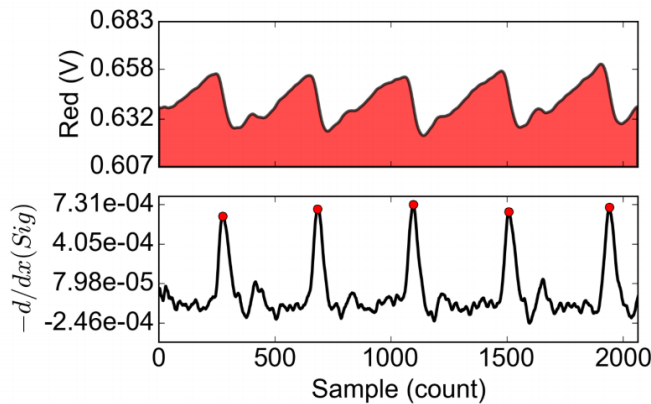
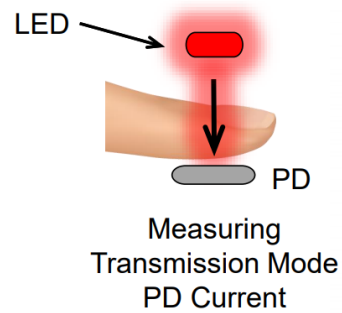


Figure 4.7 Calibration curves for pulse oximeters: the solid line is the theoretical curve by Beer's law and the dashed line is the empirical curve. The difference between these curves is due mainly to light scattering effects. This empirical calibration curve is derived by a second order polynomial.

$$R_{os} = \frac{\frac{Red_{ac}}{Red_{dc}}}{\frac{NIR_{ac}}{NIR_{dc}}}$$

R_{os} comes from hardware

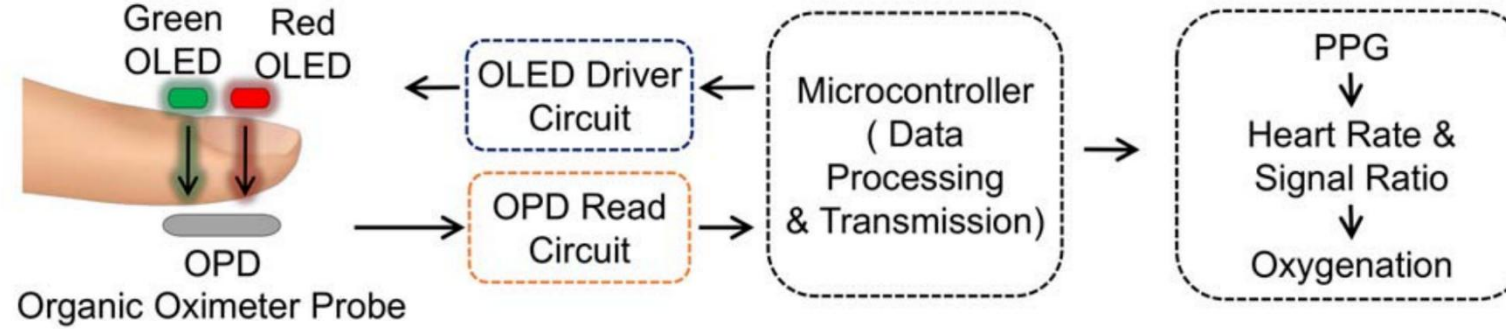
Calculating oxygen saturation



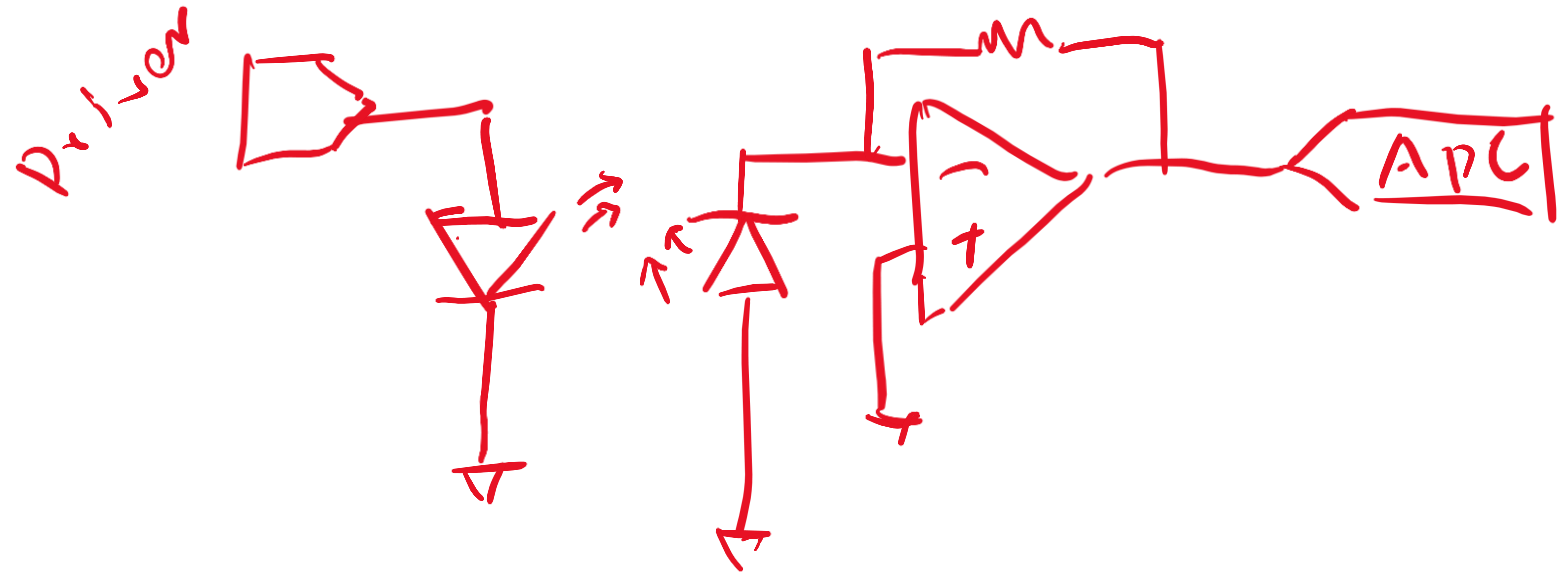
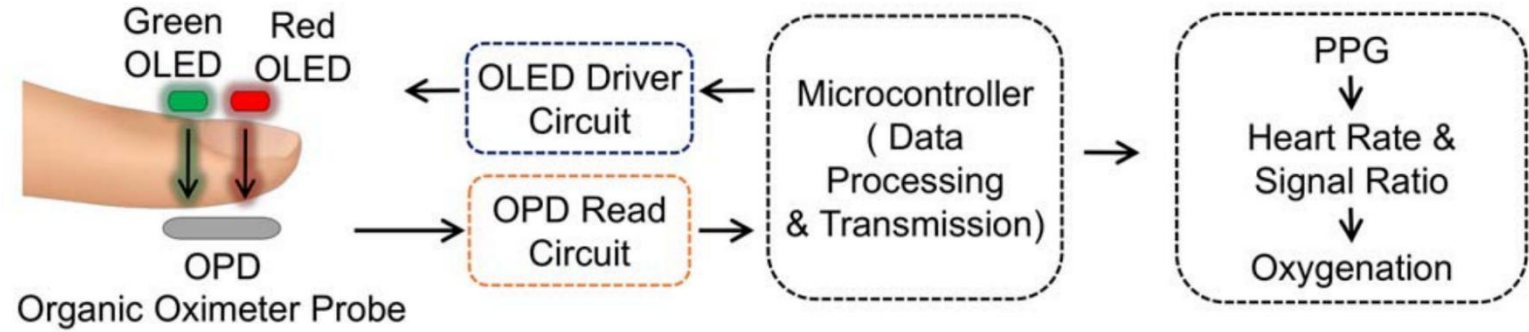
$$R = \frac{AC_{rd}/DC_{rd}}{AC_{ir}/DC_{ir}}$$

$$R = \frac{20mV/625mV}{20mV/330mV} = .528$$

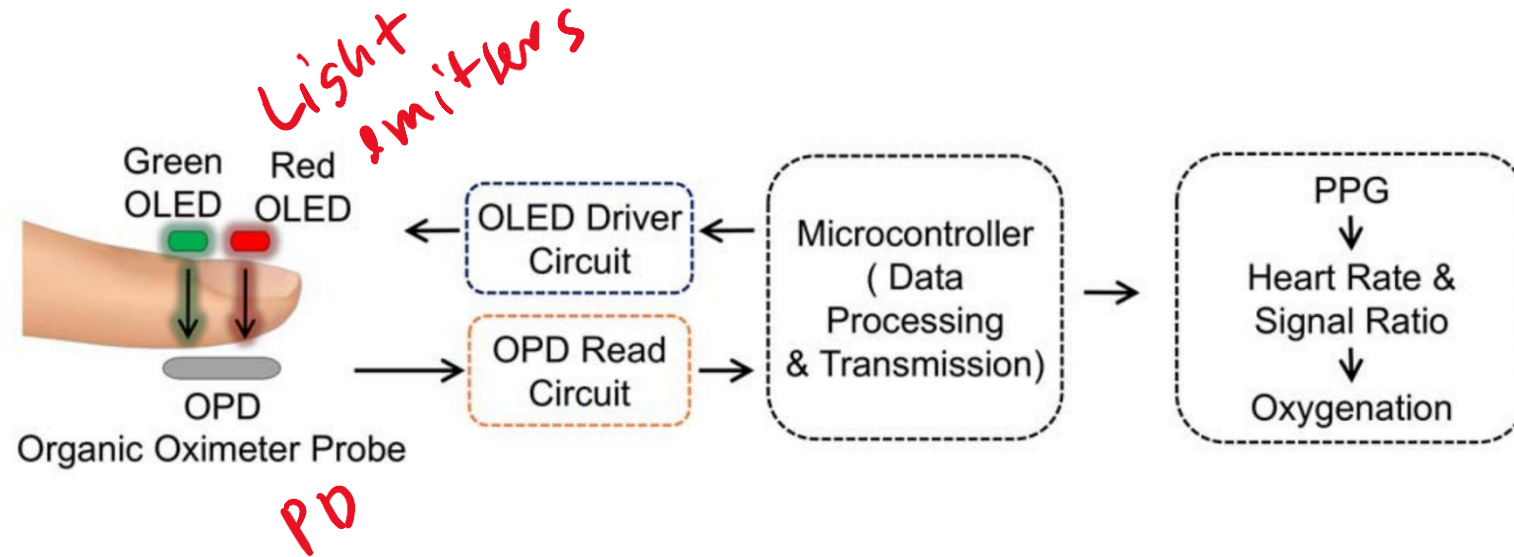
Oximeter readout circuit



Oximeter readout circuit



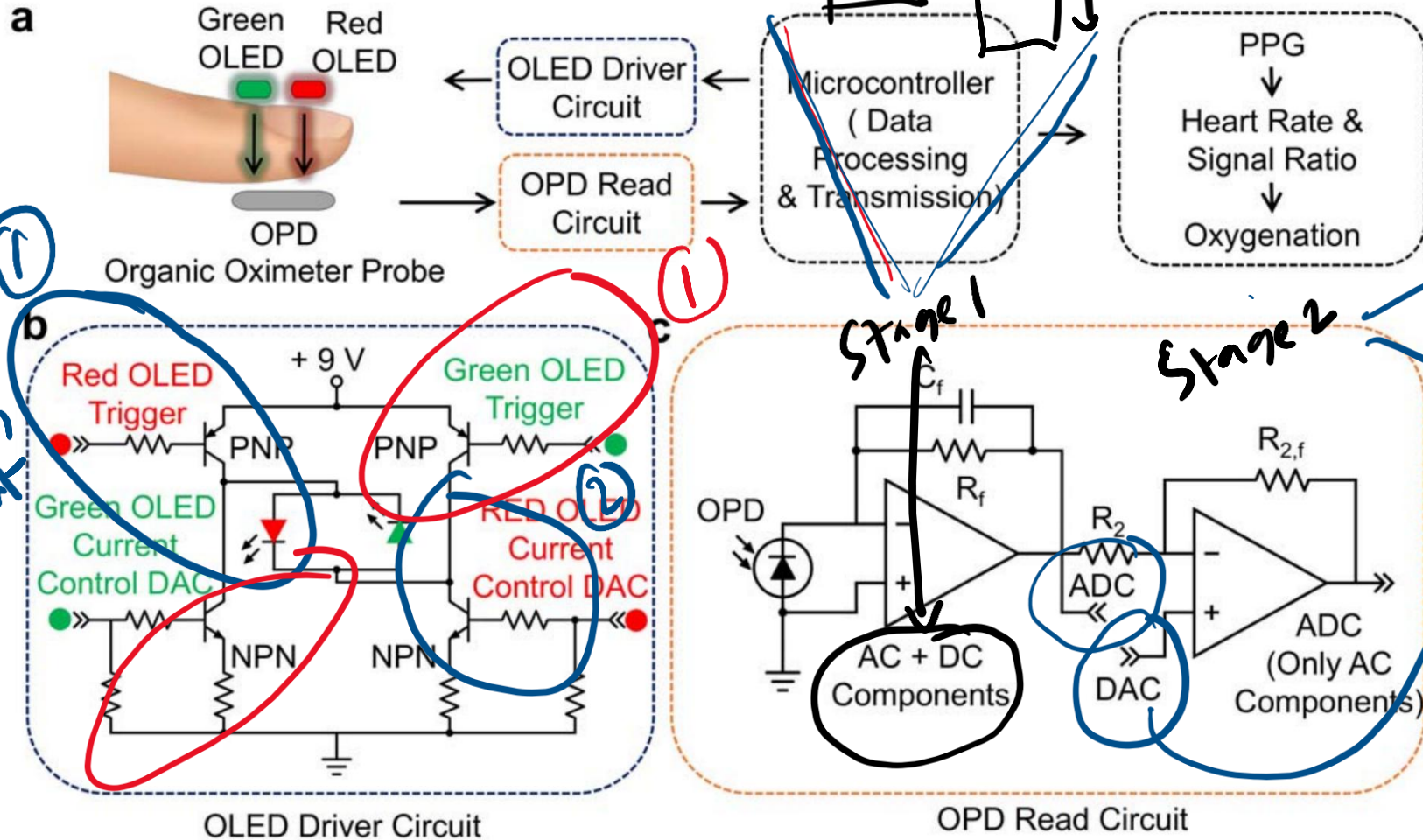
Oximeter readout circuit



LED driver → (1) Trigger to turn on
(2) Control LED current to change light intensity

PD circuit (1) Amplify (AC+DC)
(2) Measure and get rid off DC
(3) Amplify AC

Oximeter readout circuit



① Trigger
② Controlling current

Stage 1

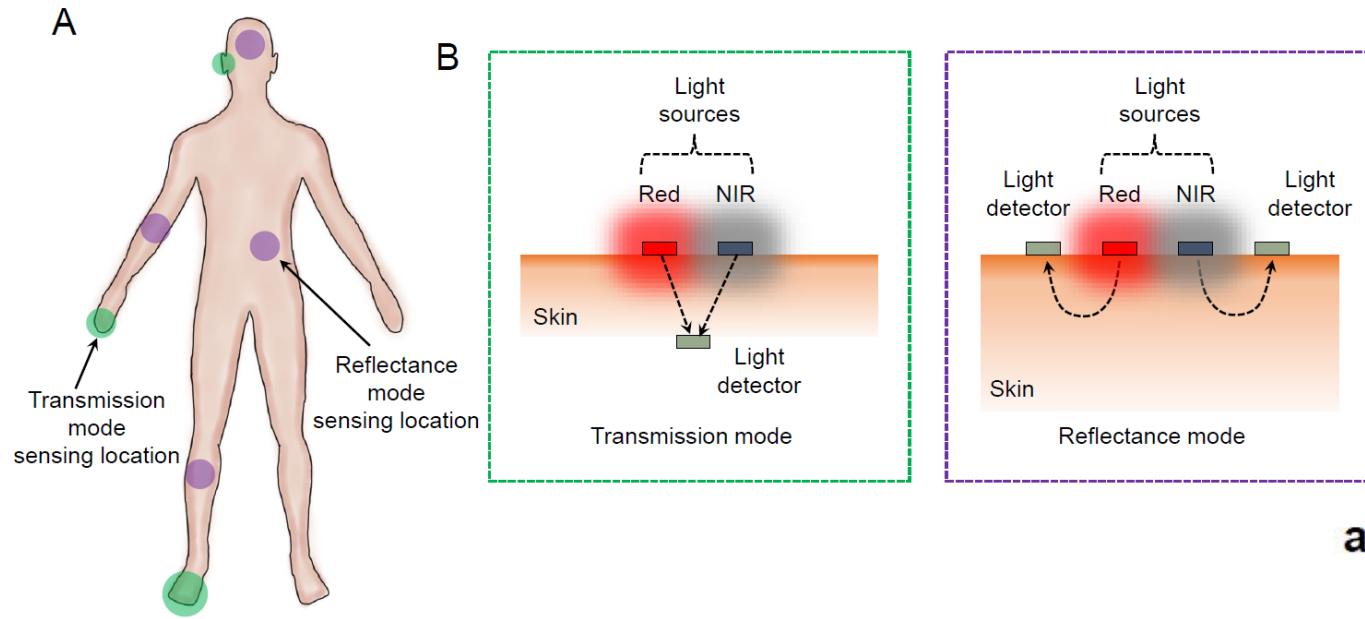
Stage 2

AC

get rid off DC amplify AC

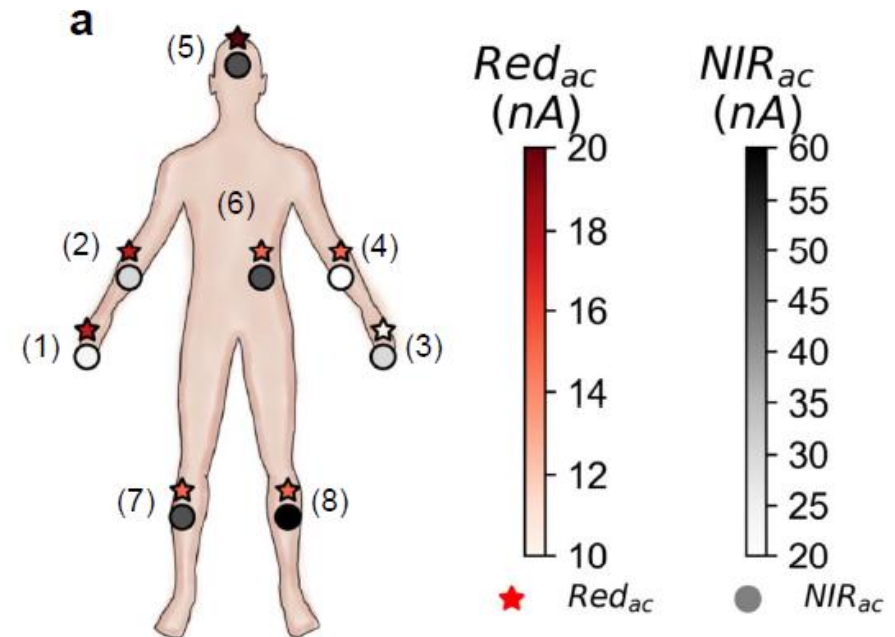
$$V_{out} = (V_{+} - V_{-}) \cdot A$$

Transmission vs. reflectance oximetry



- Transmission-mode pulse oximetry is limited only to tissues that can be transilluminated, such as the earlobes and the fingers.
- If reflected light is used as the signal, the sensor can be used beyond the conventional sensing locations.

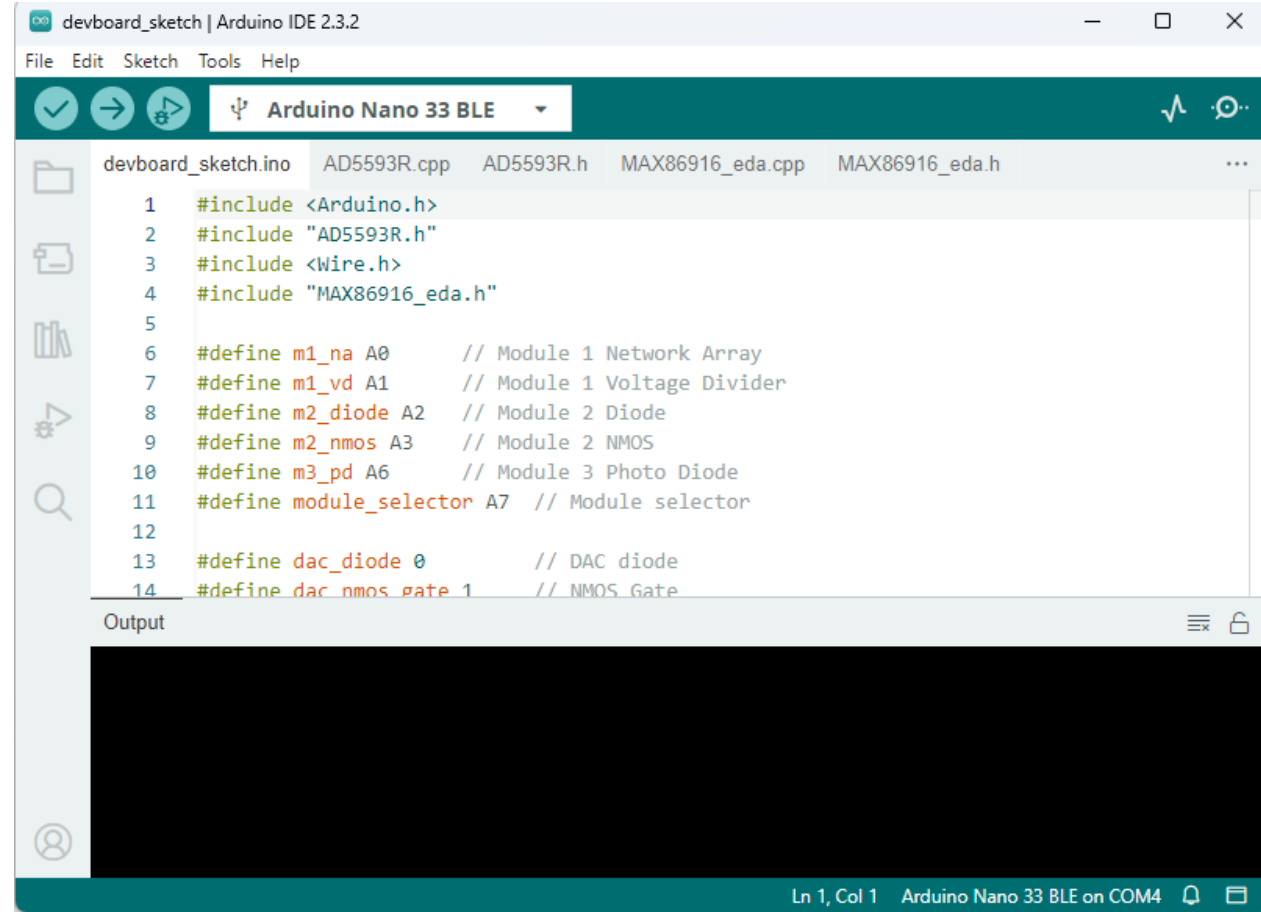
- AC signal is the highest at the forehead for both Red and NIR channels.
- Arms provide mid-range AC amplitude, while signal strength is low in the legs and chest area.
- Forehead is the best location for reflectance pulse oximetry.



Collect PPG signal

Downloads > ppg_module > ppg_module >		
  Sort ▾  View ▾ 		
☐ Name	Date modified ▾	Type
Today		
 devboard_sketch	9/29/2024 4:20 AM	File folder
 notebook	9/29/2024 4:20 AM	File folder

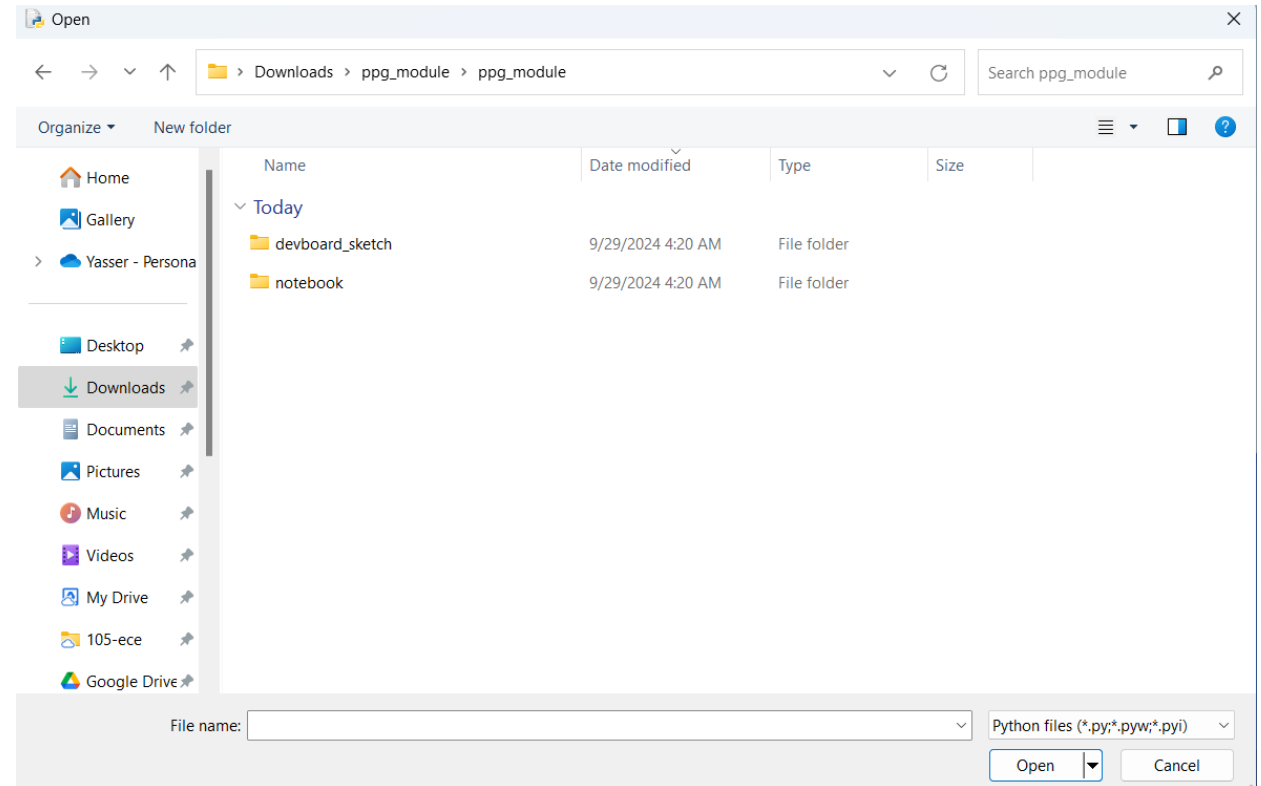
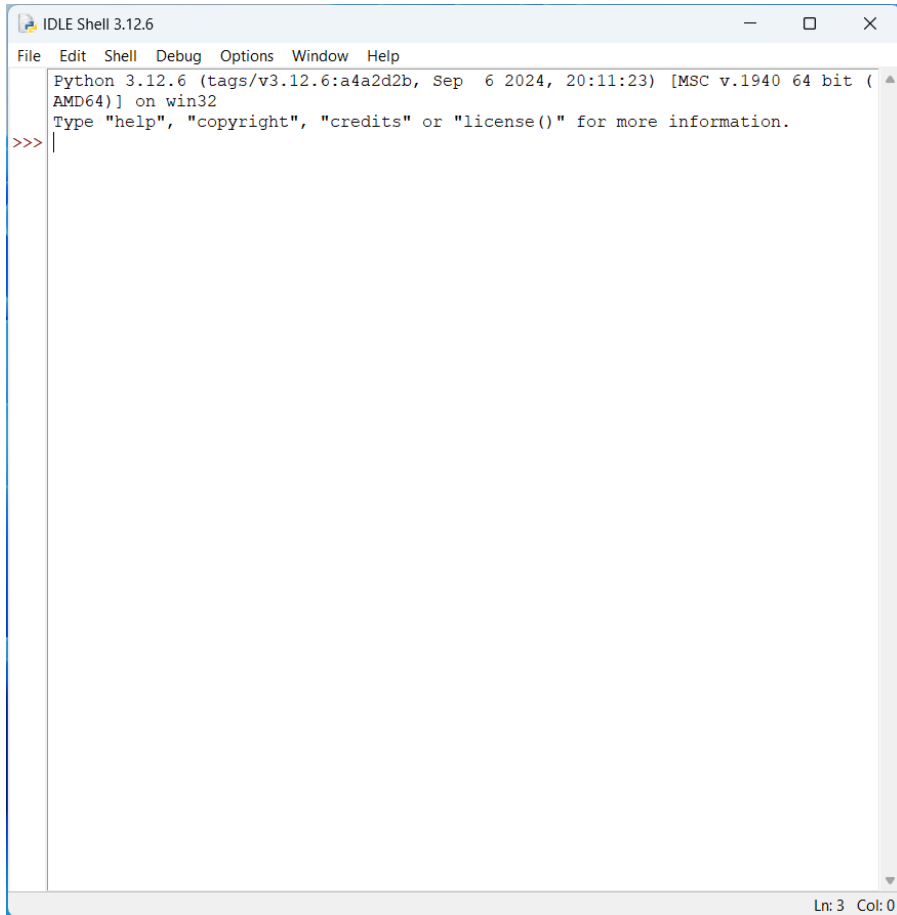
Upload and code



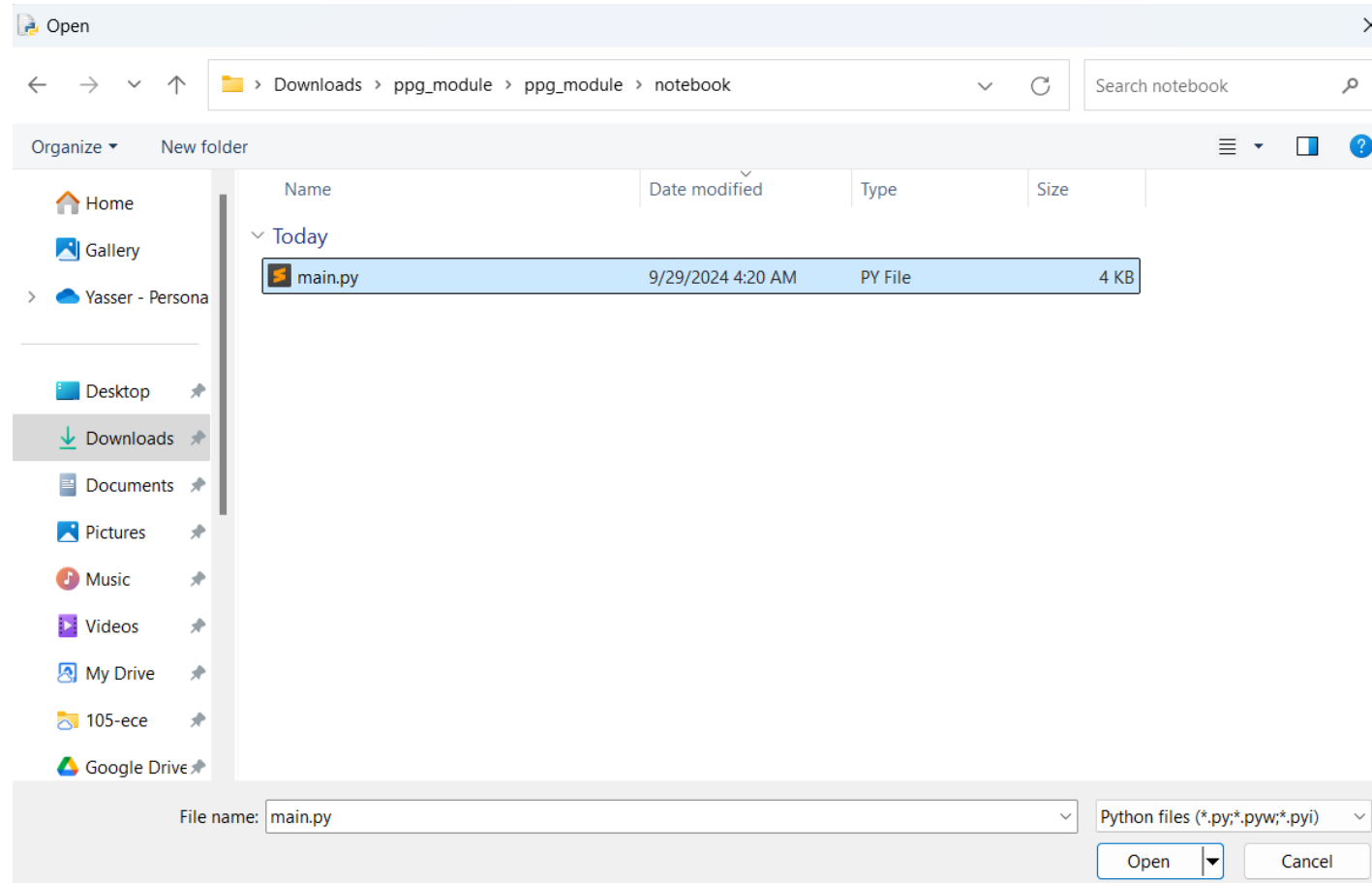
```
devboard_sketch | Arduino IDE 2.3.2
File Edit Sketch Tools Help
[Check] [Next] [Upload] [Serial Monitor] [Serial Plotter] [Tools] [Port] [Board]
devboard_sketch.ino AD5593R.cpp AD5593R.h MAX86916_eda.cpp MAX86916_eda.h ...
1 #include <Arduino.h>
2 #include "AD5593R.h"
3 #include <Wire.h>
4 #include "MAX86916_eda.h"
5
6 #define m1_na A0 // Module 1 Network Array
7 #define m1_vd A1 // Module 1 Voltage Divider
8 #define m2_diode A2 // Module 2 Diode
9 #define m2_nmos A3 // Module 2 NMOS
10 #define m3_pd A6 // Module 3 Photo Diode
11 #define module_selector A7 // Module selector
12
13 #define dac_diode 0 // DAC diode
14 #define dac_nmos_gate 1 // NMOS Gate

Output
Ln 1, Col 1 Arduino Nano 33 BLE on COM4
```

Open python IDLE – open python file



Open main.py



Run module

```
main.py - C:\Users\tasif\Downloads\ppg_module\ppg_module\notebook\main.py (3.12.6)
File Edit Format Run Options Window Help
import serial
import matplotlib.pyplot as plt
import matplotlib.animation as animation
from matplotlib.widgets import Button
import csv
import datetime
import time
import collections

# Prompt user to enter the serial port number
port = input("Enter the serial port (e.g., COM5 or /dev/ttyUSB0): ").strip()

# Set up serial connection with the user-provided port
ser = serial.Serial(port, 115200, timeout=1)

time.sleep(2)

print("="*50)
print("Note:")
print("Ensure Module 3 is activated before proceeding.")
print("Activate the PPG sensor on the EE105 devboard using the command 'm3ppg'."
print("="*50)

print("Type your command:")
command = input("> ").strip()
ser.write((command + '\n').encode())
print("-"*50)

# Data storage for RED and IR values
red_data = collections.deque(maxlen=1000) # Keep the last 1000 RED data points
ir_data = collections.deque(maxlen=1000) # Keep the last 1000 IR data points

# Data capture state
capturing = False

# Set up live plotting with two subplots
fig, (ax1, ax2) = plt.subplots(2, 1, figsize=(12, 8)) # 2 rows, 1 column, adjust
red_line, = ax1.plot([], [], lw=2, label='RED', color='red')
ir_line, = ax2.plot([], [], lw=2, label='IR', color='blue')
ax1.set_xlim(0, 1000) # Display the last 1000 data points on RED subplot
Ln: 1 Col: 0
```

```
main.py - C:\Users\tasif\Downloads\ppg_module\ppg_module\notebook\main.py (3.12.6)
File Edit Format Run Options Window Help
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import matplotlib.pyplot as plt
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print("Activate the PPG sensor on the EE105 devboard using the command 'm3ppg'."
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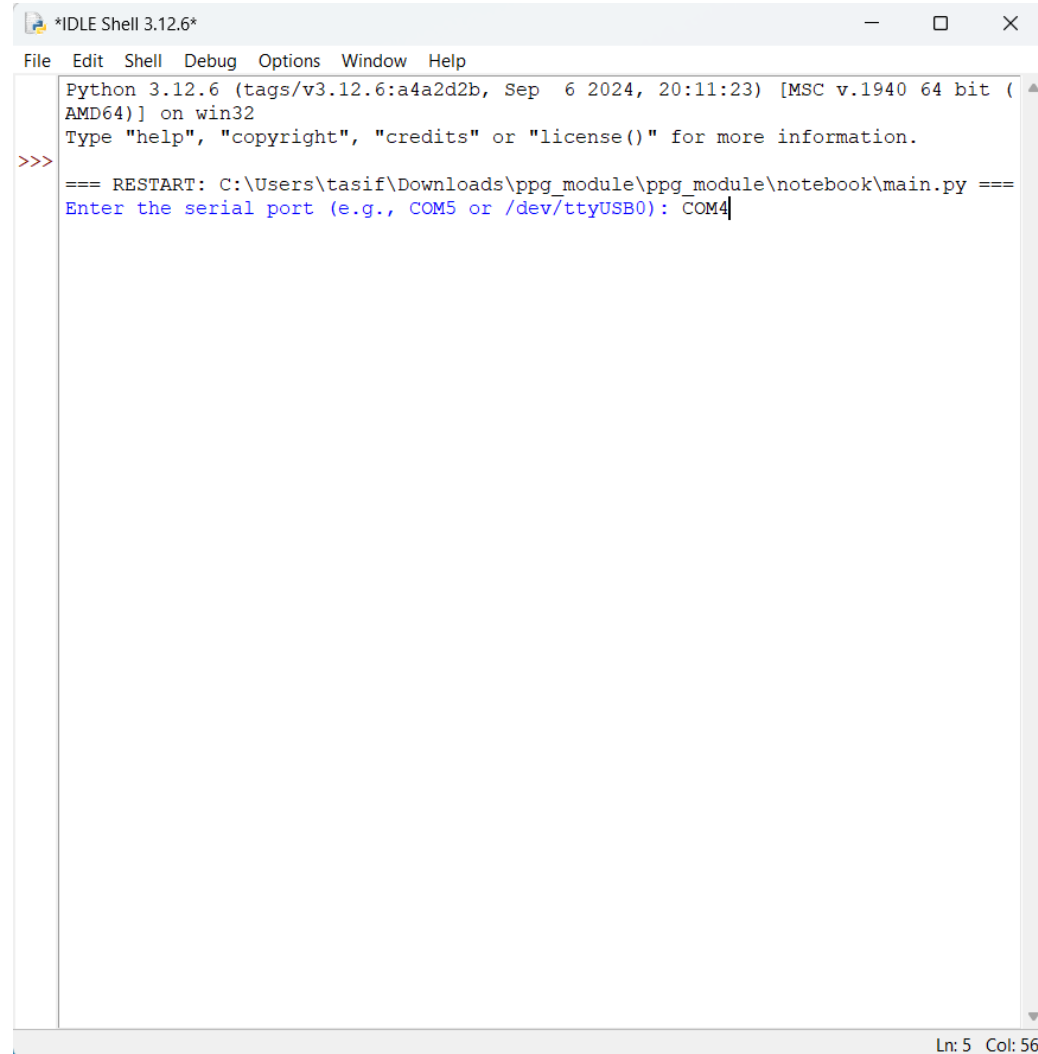
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Ln: 1 Col: 0
```

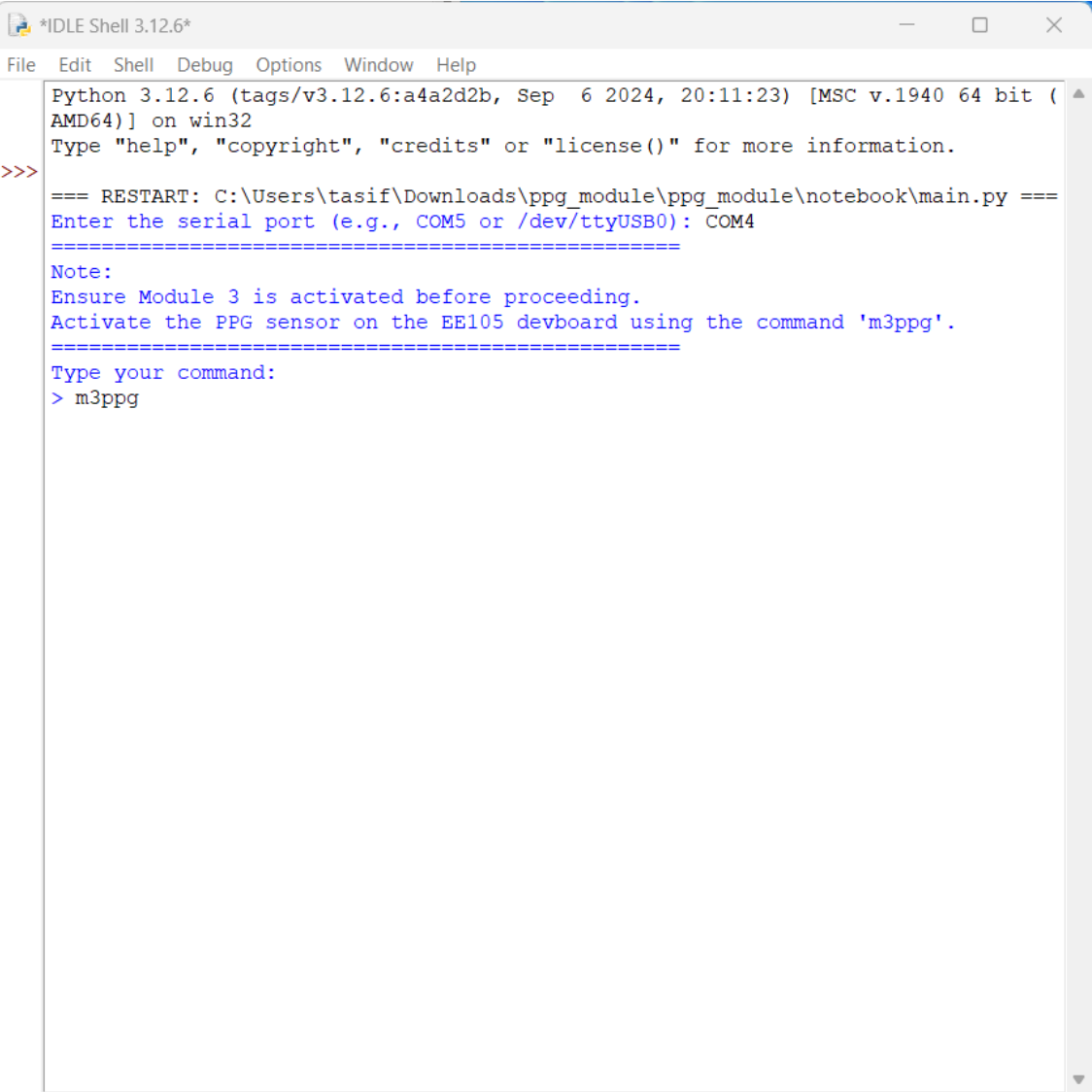
Type in which port – check where your Ard is connected



```
*IDLE Shell 3.12.6*
File Edit Shell Debug Options Window Help
Python 3.12.6 (tags/v3.12.6:a4a2d2b, Sep 6 2024, 20:11:23) [MSC v.1940 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
=== RESTART: C:\Users\tasif\Downloads\ppg_module\ppg_module\notebook\main.py ===
Enter the serial port (e.g., COM5 or /dev/ttyUSB0): COM4
```

Ln: 5 Col: 56

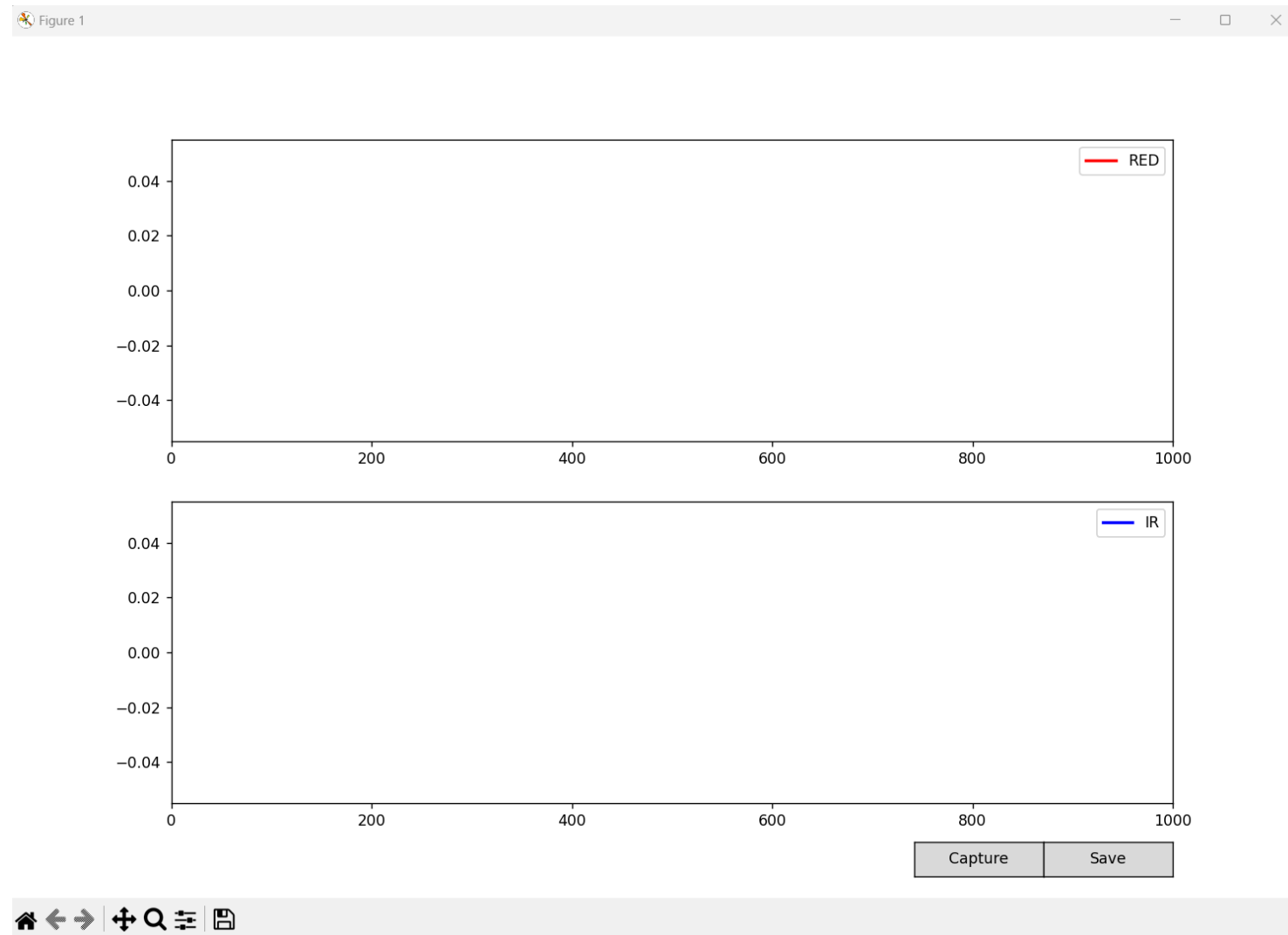
Command – m3ppg



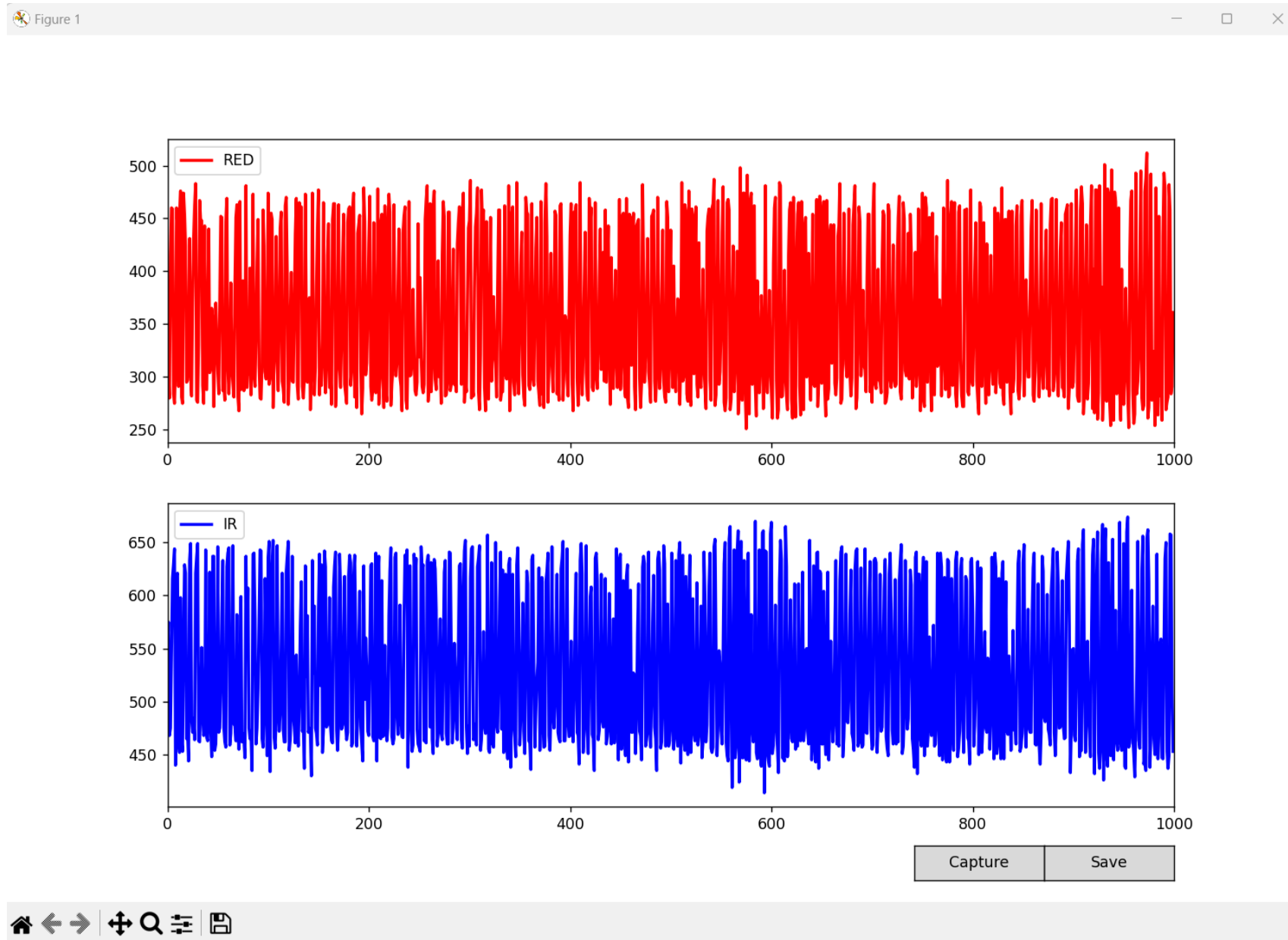
```
*IDLE Shell 3.12.6*
File Edit Shell Debug Options Window Help
Python 3.12.6 (tags/v3.12.6:a4a2d2b, Sep  6 2024, 20:11:23) [MSC v.1940 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
=== RESTART: C:\Users\tasif\Downloads\ppg_module\ppg_module\notebook\main.py ===
Enter the serial port (e.g., COM5 or /dev/ttyUSB0): COM4
=====
Note:
Ensure Module 3 is activated before proceeding.
Activate the PPG sensor on the EE105 devboard using the command 'm3ppg'.
=====
Type your command:
> m3ppg
```

Ln: 12 Col: 7

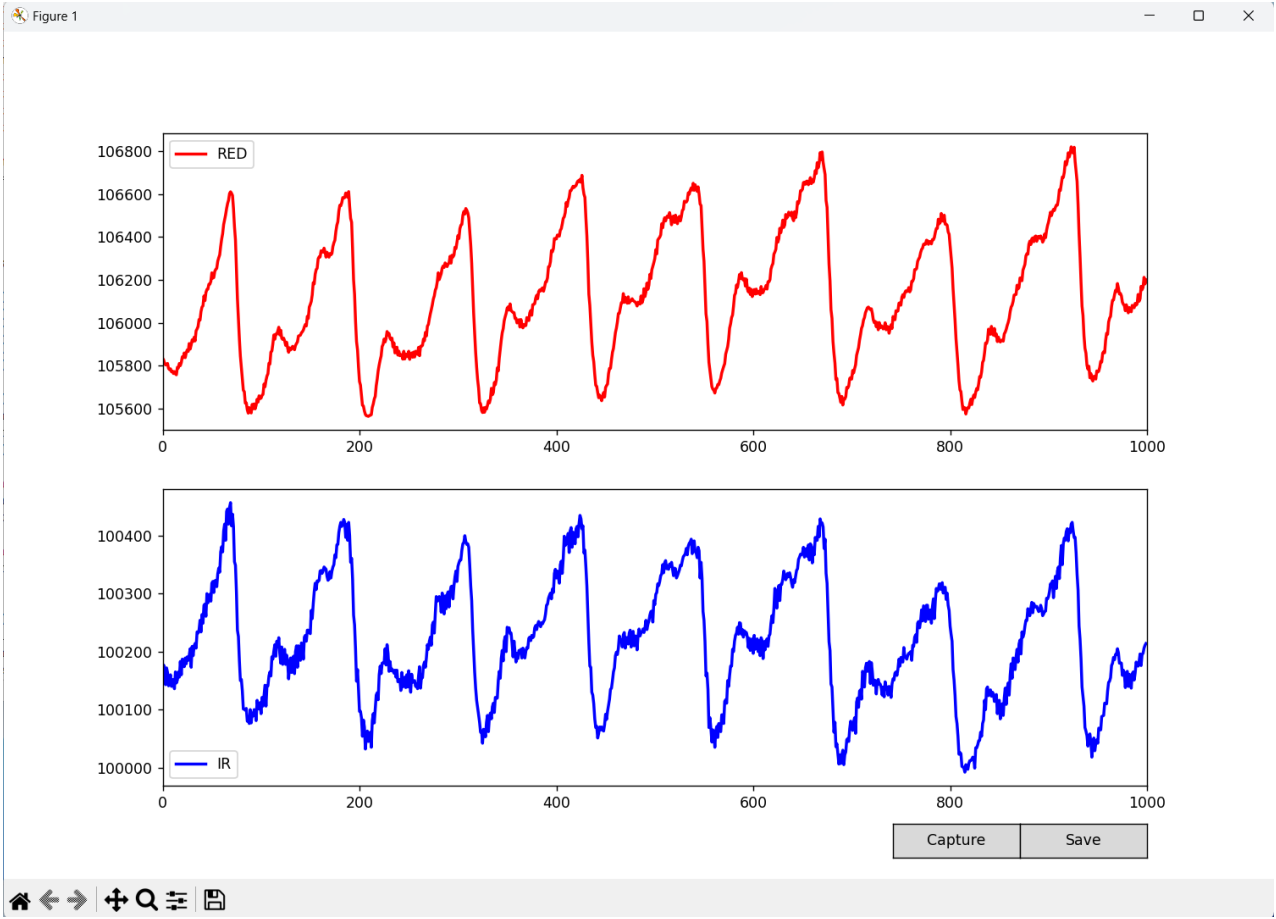
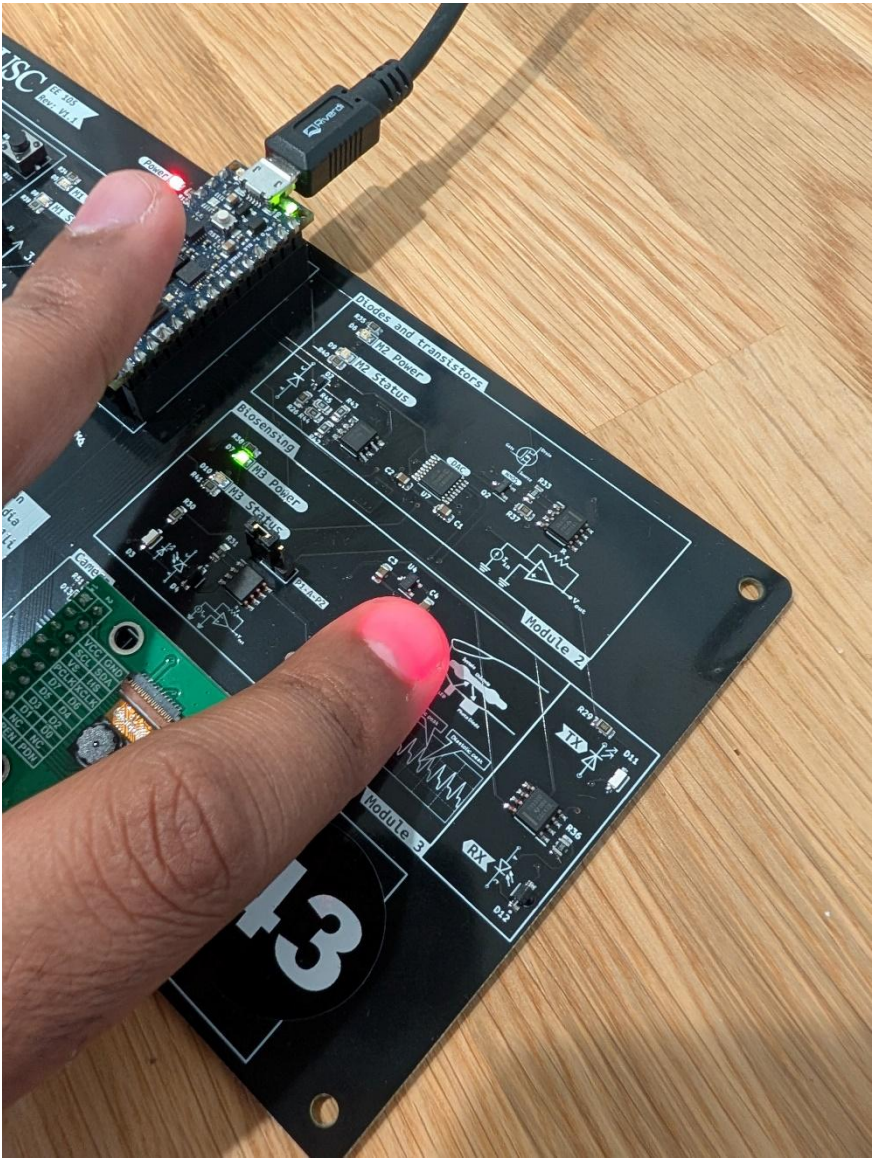
Opens a recording window



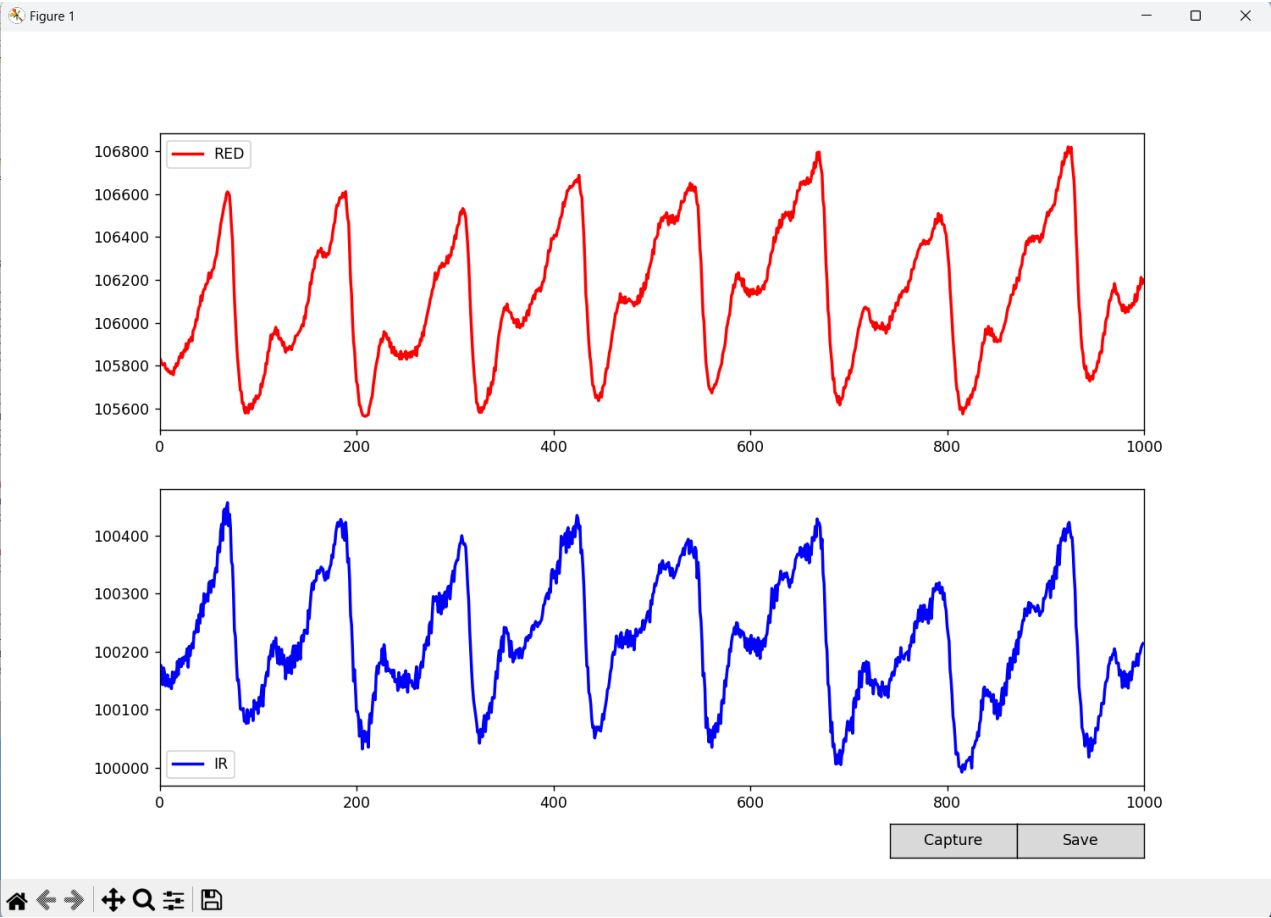
Press capture



Keep your finger gently on the sensor



Click save – a new file will be saved



Select file

Downloads > ppg_module > ppg_module > notebook

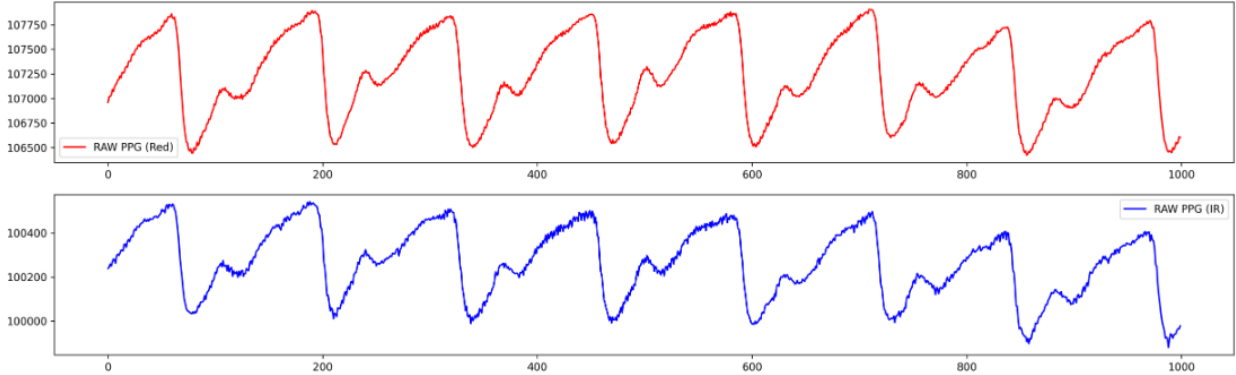
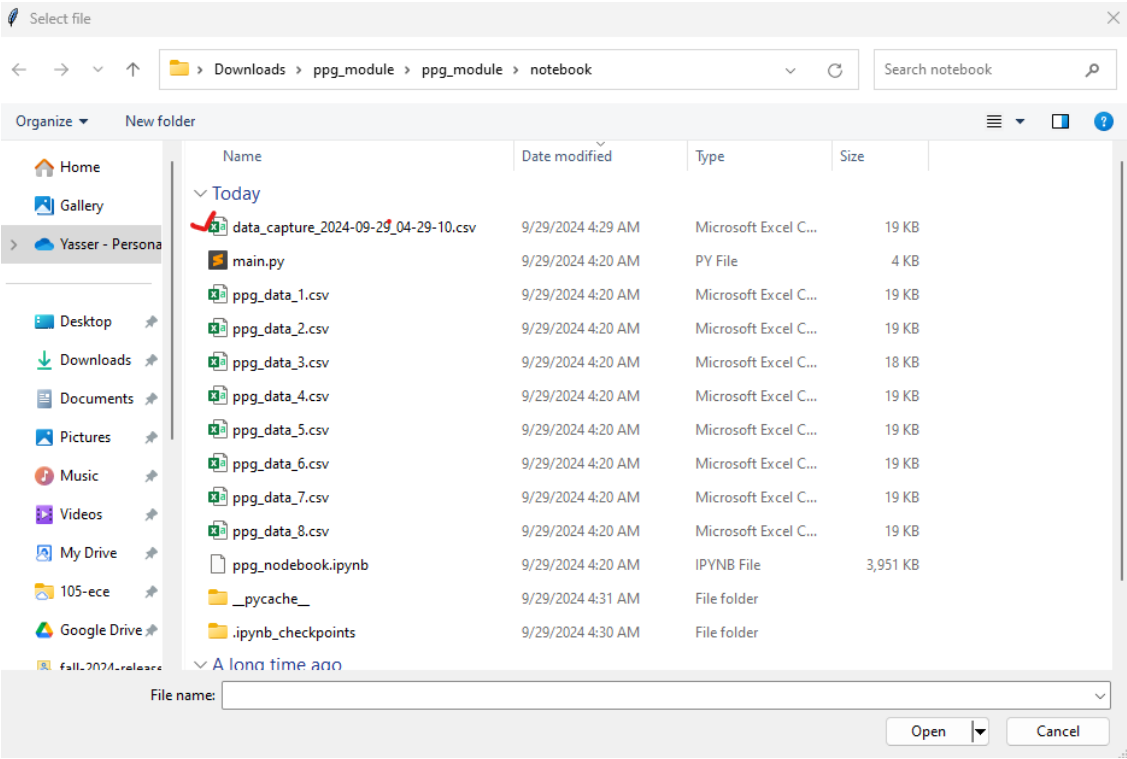
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Open jupyter notebook select the new csv file – check the raw data on notebook



Pulse oximeter basics

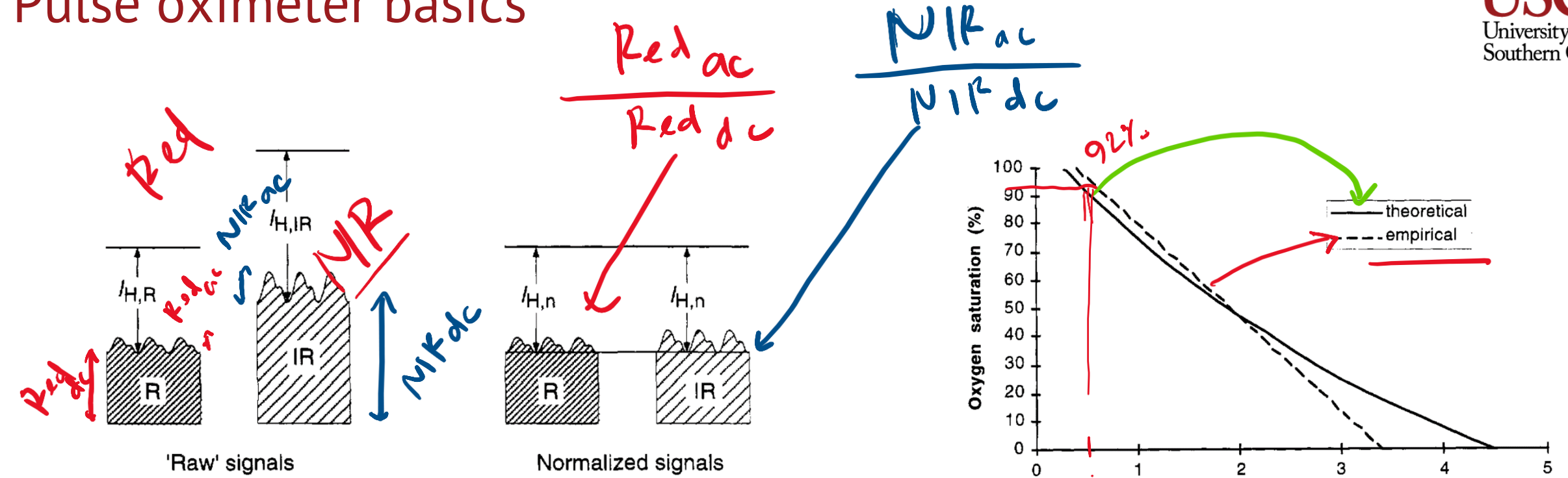
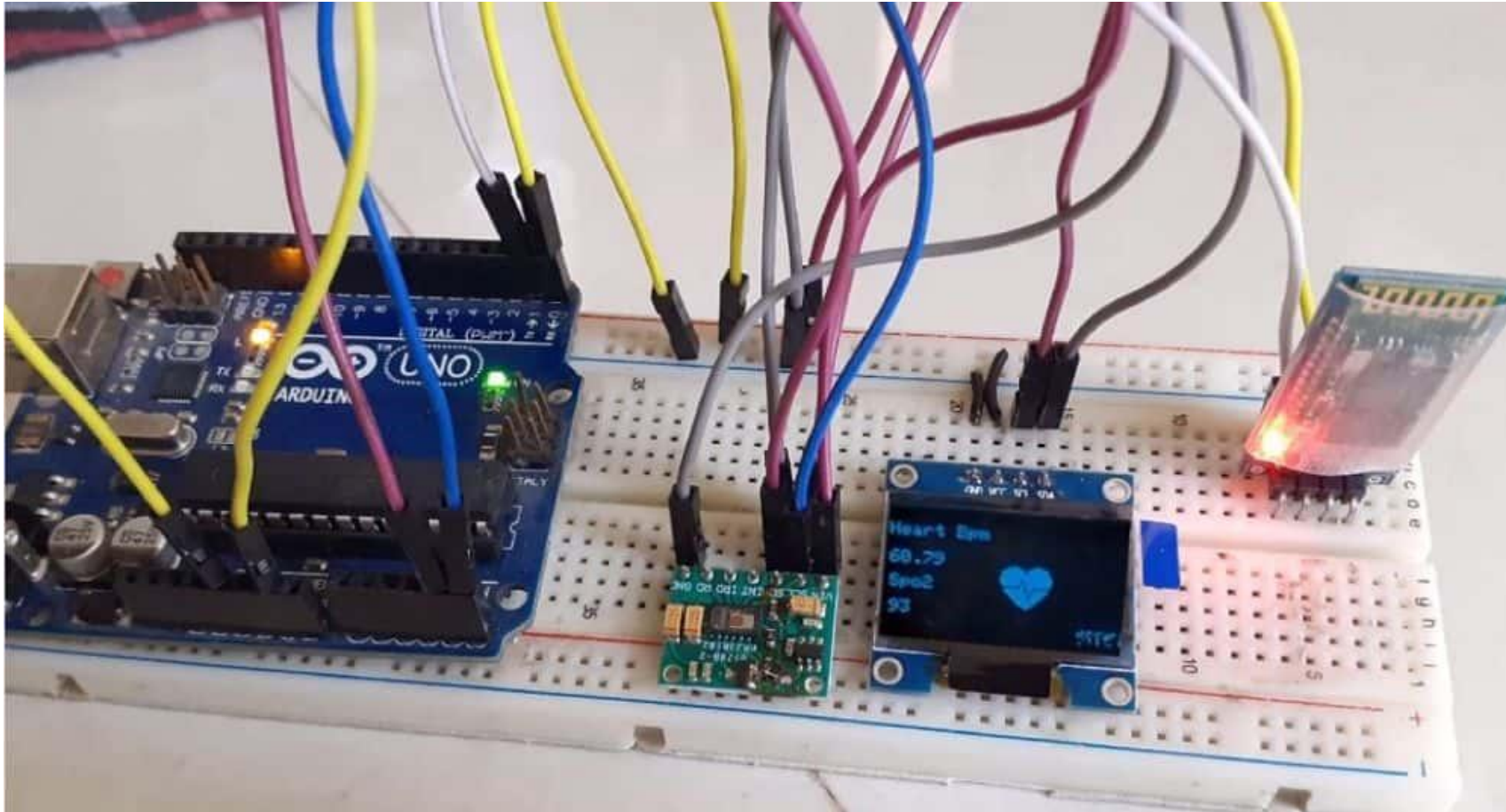


Figure 4.6 The normalization of the signals. The transmitted light from the red LED (R) and from the infrared LED (IR) is divided by its individual DC component. Thus, both normalized light intensities have the same magnitude during diastole. The normalized signals determine the basis for the calculation of the arterial oxygen saturation.

Figure 4.7 Calibration curves for pulse oximeters: the solid line is the theoretical curve by Beer's law and the dashed line is the empirical curve. The difference between these curves is due mainly to light scattering effects. This empirical calibration curve is derived by a second order polynomial.

Oximeter prototype



Click [here](#) for production status of specific part numbers.

MAX30102

High-Sensitivity Pulse Oximeter and Heart-Rate Sensor for Wearable Health

General Description

The MAX30102 is an integrated pulse oximetry and heart-rate monitor module. It includes internal LEDs, photodetectors, optical elements, and low-noise electronics with ambient light rejection. The MAX30102 provides a complete system solution to ease the design-in process for mobile and wearable devices.

The MAX30102 operates on a single 1.8V power supply and a separate 3.3V power supply for the internal LEDs. Communication is through a standard I²C-compatible interface. The module can be shut down through software with zero standby current, allowing the power rails to remain powered at all times.

Applications

- Wearable Devices
- Fitness Assistant Devices
- Smartphones
- Tablets

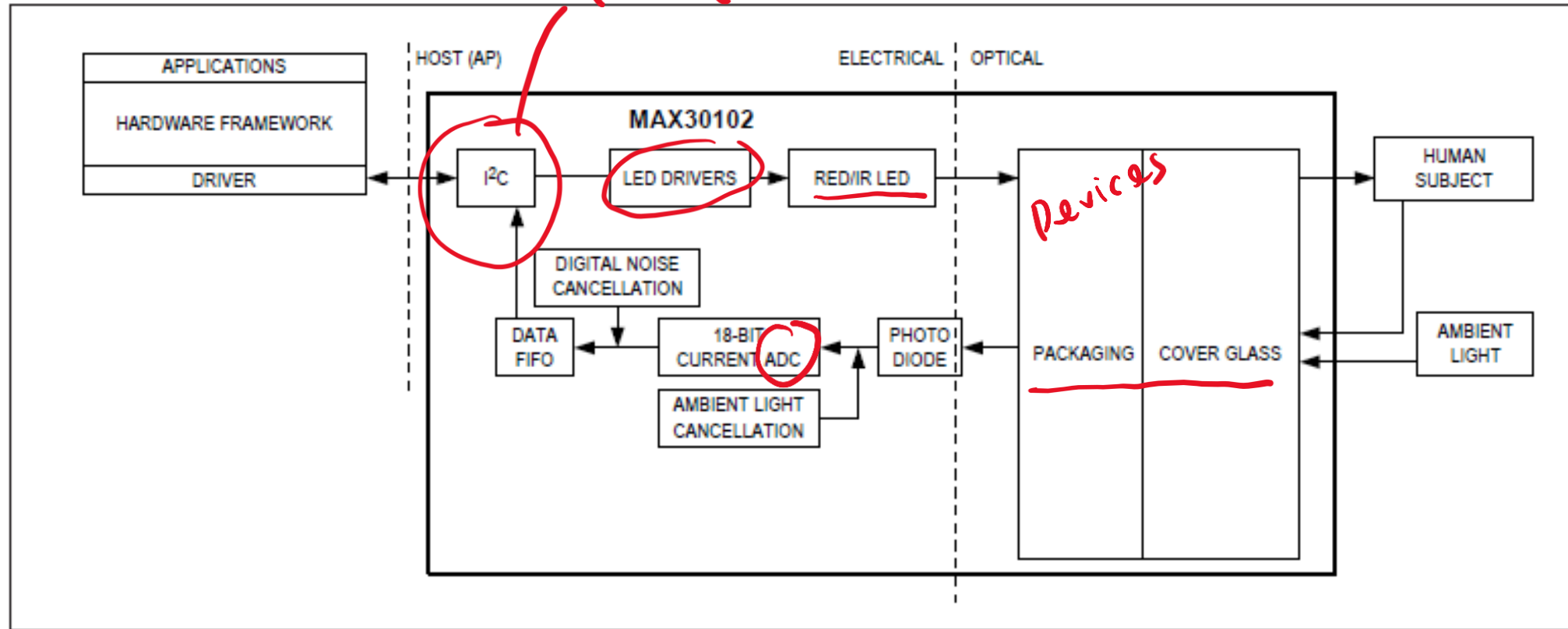
Benefits and Features

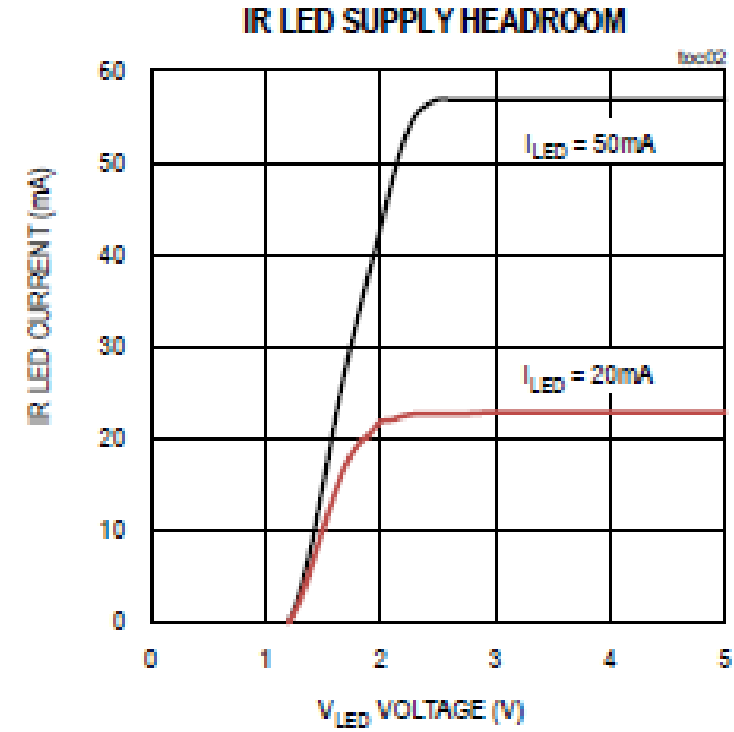
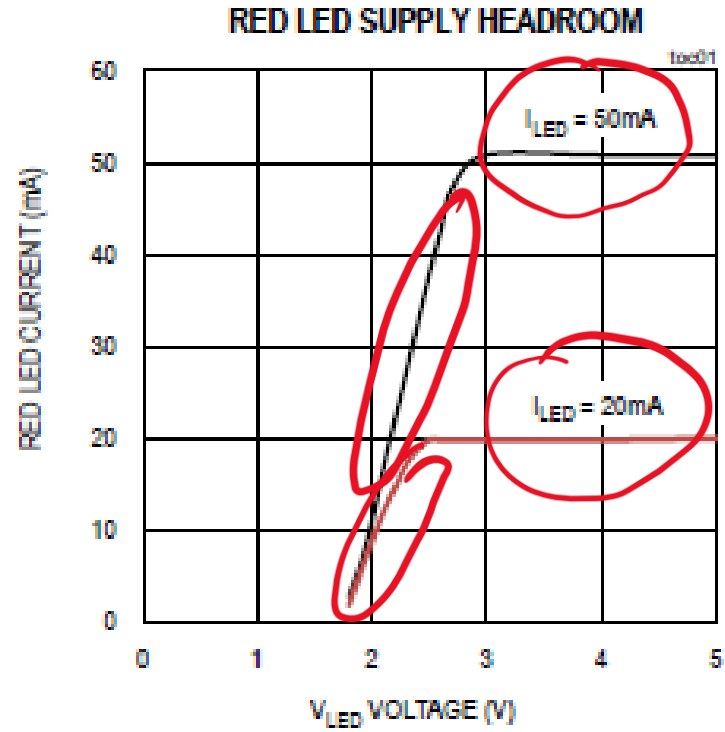
- Heart-Rate Monitor and Pulse Oximeter Sensor in LED Reflective Solution
- Tiny 5.6mm x 3.3mm x 1.55mm 14-Pin Optical Module
 - Integrated Cover Glass for Optimal, Robust Performance
- Ultra-Low Power Operation for Mobile Devices
 - Programmable Sample Rate and LED Current for Power Savings
 - Low-Power Heart-Rate Monitor (< 1mW)
 - Ultra-Low Shutdown Current (0.7μA, typ)
- Fast Data Output Capability
 - High Sample Rates
- Robust Motion Artifact Resilience
 - High SNR
- -40°C to +85°C Operating Temperature Range

[Ordering Information](#) appears at end of data sheet.

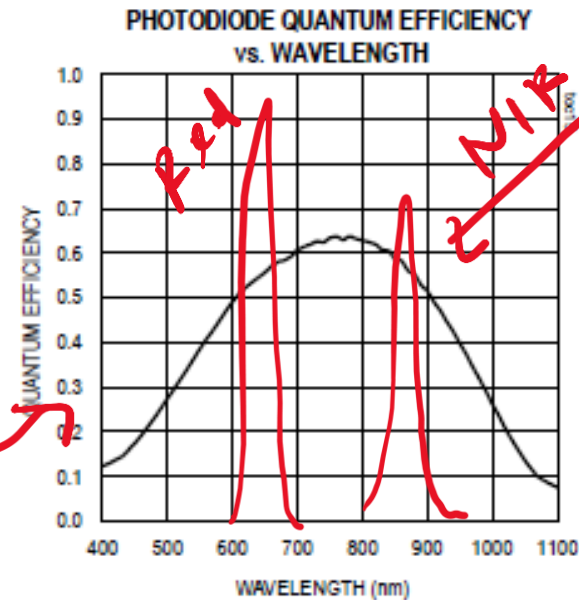
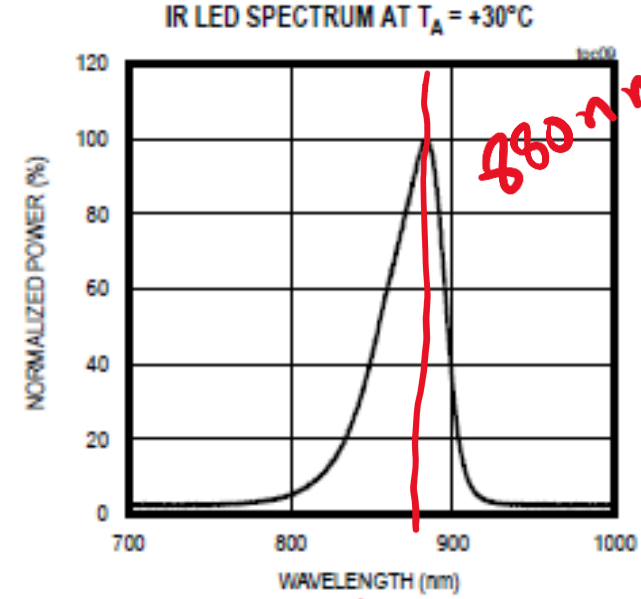
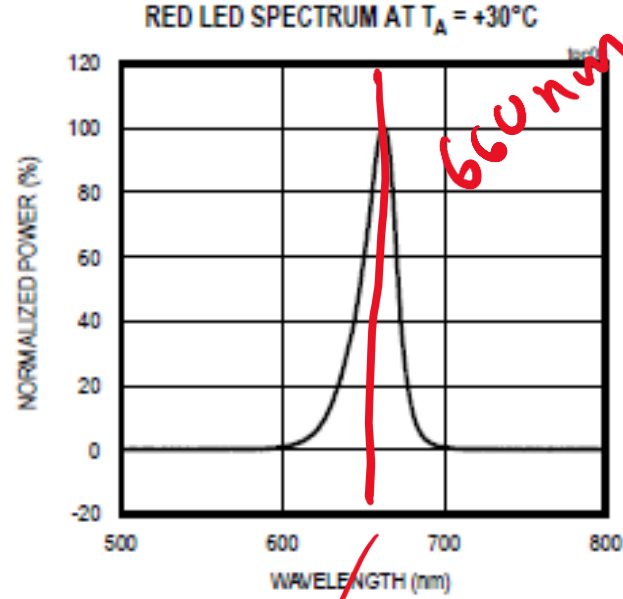


System Diagram

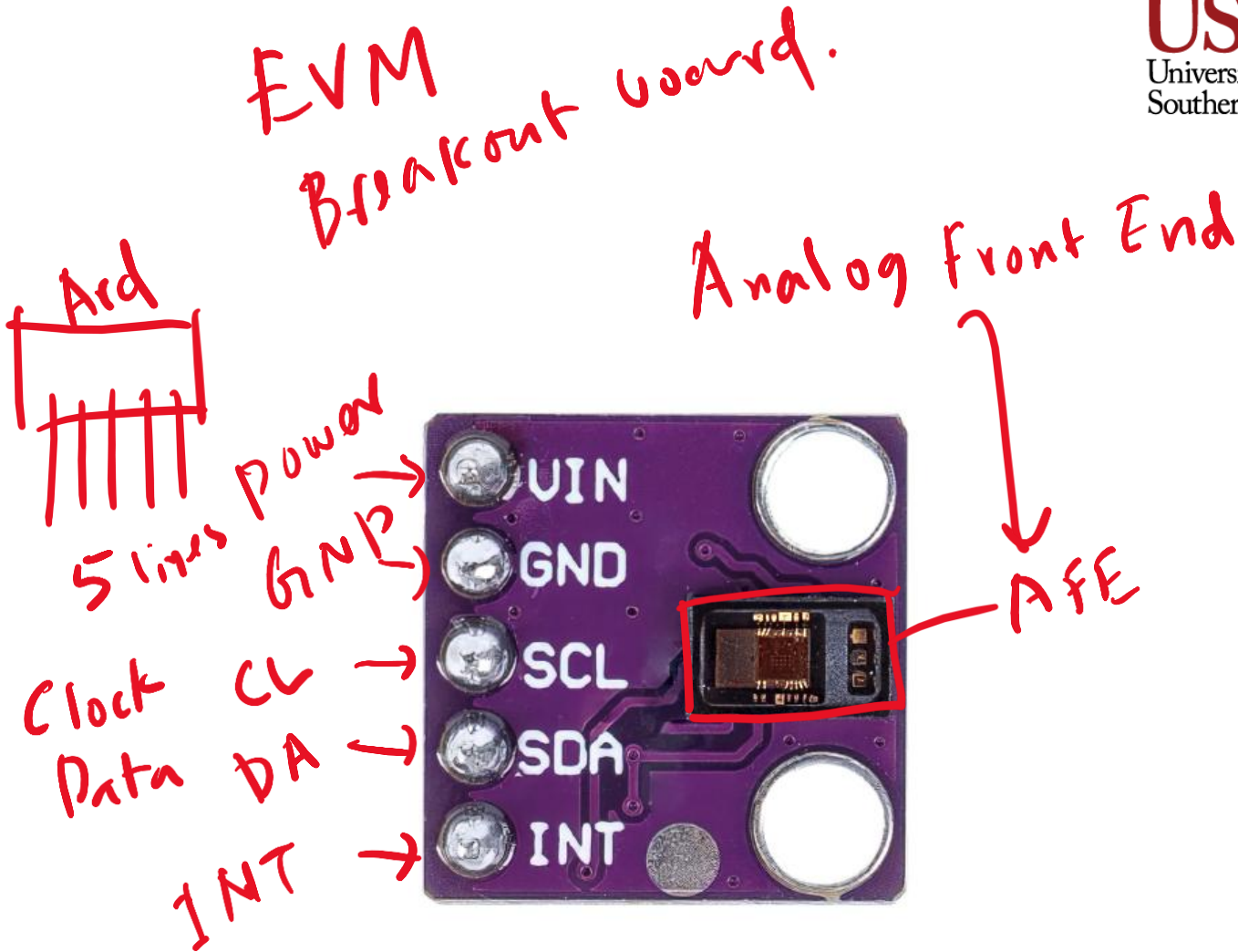
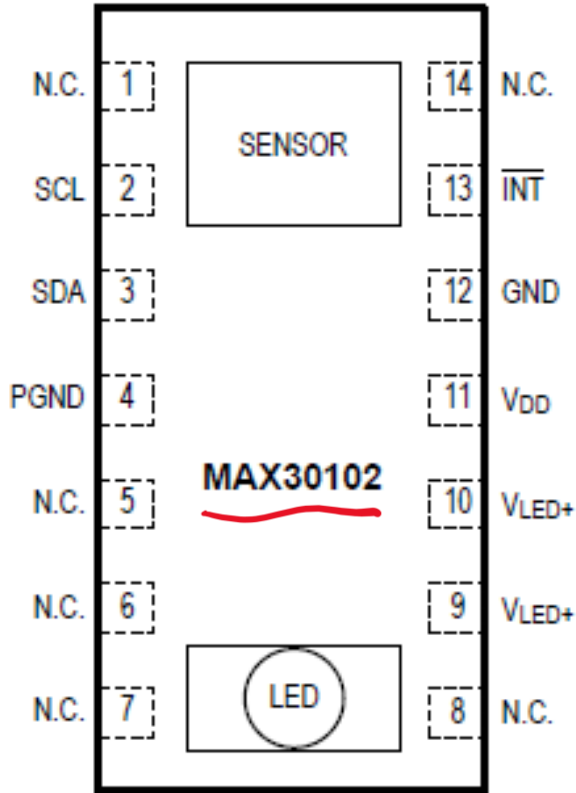




LED – PD spectrum



Connection



Connection

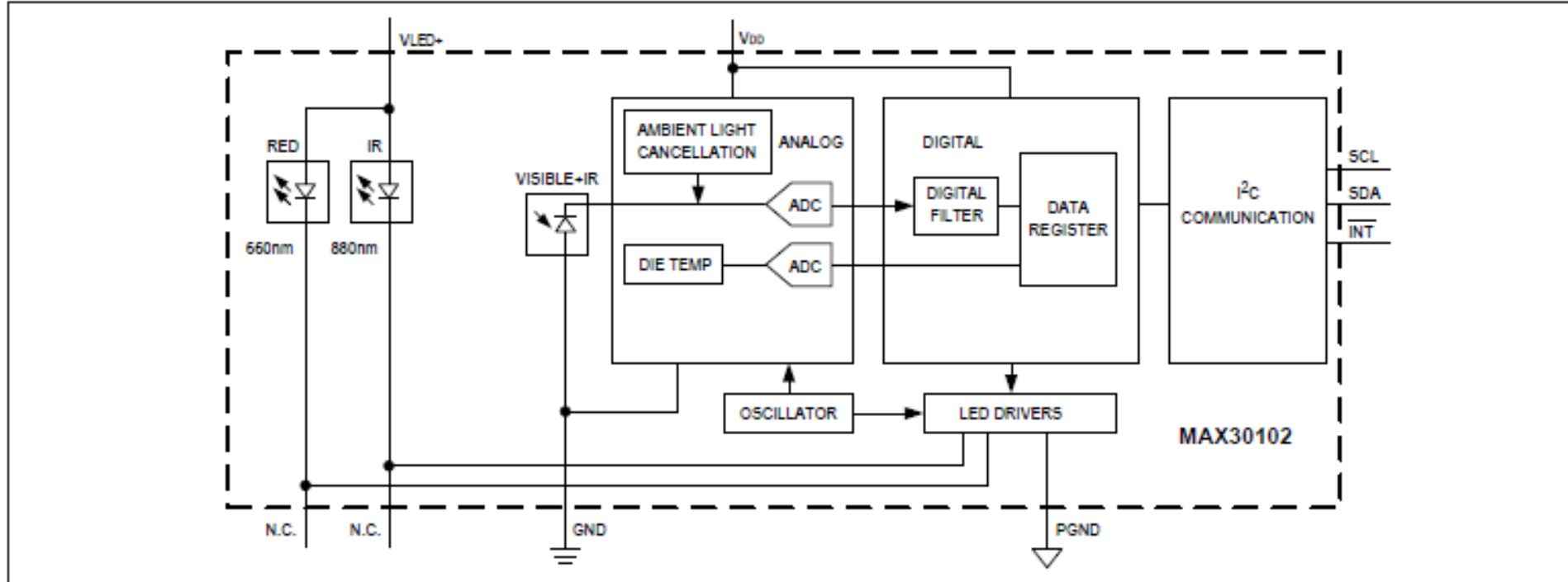
Pin Description

PIN	NAME	FUNCTION
1, 5, 6, 7, 8, 14	N.C.	No Connection. Connect to PCB pad for mechanical stability.
2	SCL	I ² C Clock Input
3	SDA	I ² C Data, Bidirectional (Open-Drain)
4	PGND	Power Ground of the LED Driver Blocks
9	V _{LED+}	LED Power Supply (anode connection). Use a bypass capacitor to PGND for best performance.
10	V _{LED+}	
11	V _{DD}	Analog Power Supply Input. Use a bypass capacitor to GND for best performance.
12	GND	
13	INT	Active-Low Interrupt (Open-Drain). Connect to an external voltage with a pullup resistor.



How does the AFE work

Functional Diagram



Data capture

Timing in SpO₂ Mode

The internal FIFO stores up to 32 samples, so that the system processor does not need to read the data after every sample. The temperature does not need to be sampled very often—once a second or every few seconds should be sufficient.

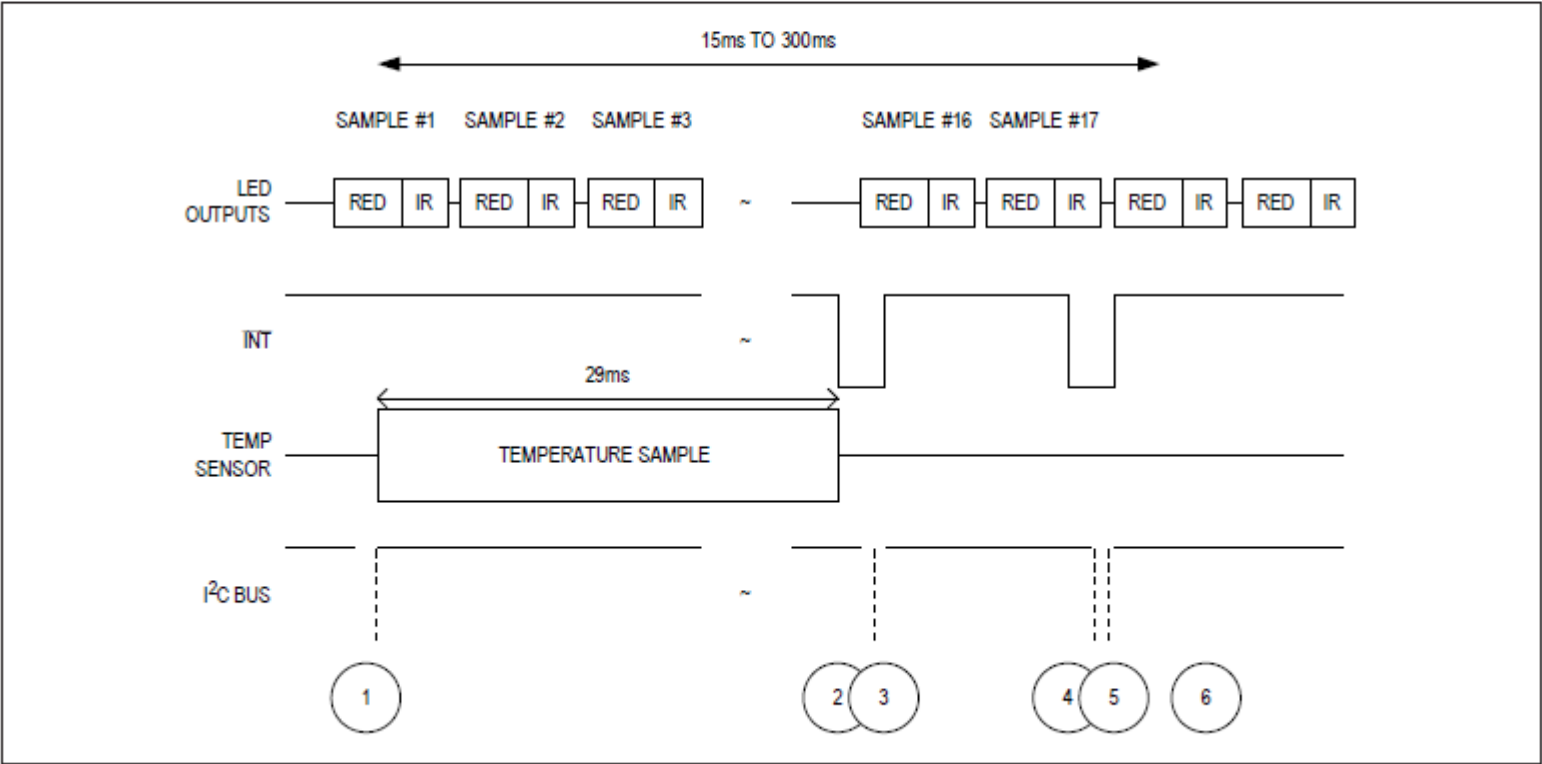


Figure 4. Timing for Data Acquisition and Communication When in SpO₂ Mode

Data capture

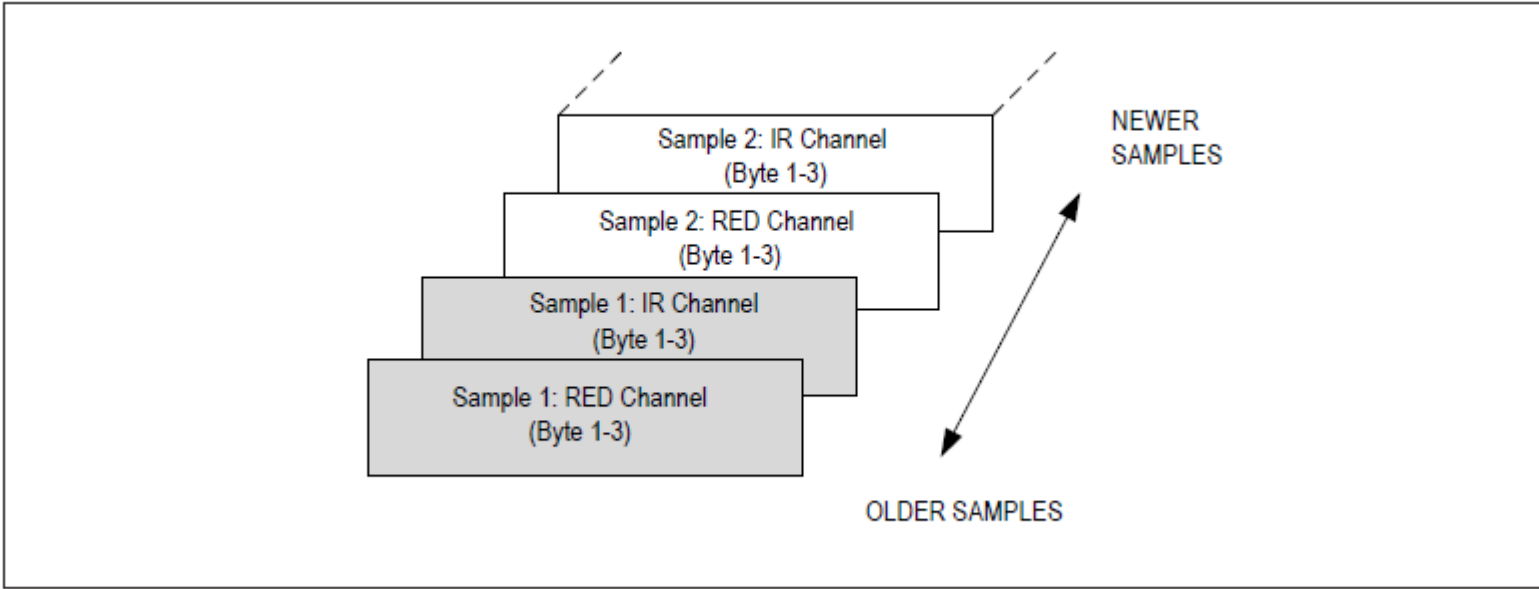


Figure 2. Graphical Representation of the FIFO Data Register. It shows IR and Red in SpO₂ Mode.

Bits 4:2: SpO₂ Sample Rate Control

These bits define the effective sampling rate with one sample consisting of one IR pulse/conversion and one Red pulse/conversion.

The sample rate and pulse width are related in that the sample rate sets an upper bound on the pulse width time. If the user selects a sample rate that is too high for the selected LED_PW setting, the highest possible sample rate is programmed instead into the register.

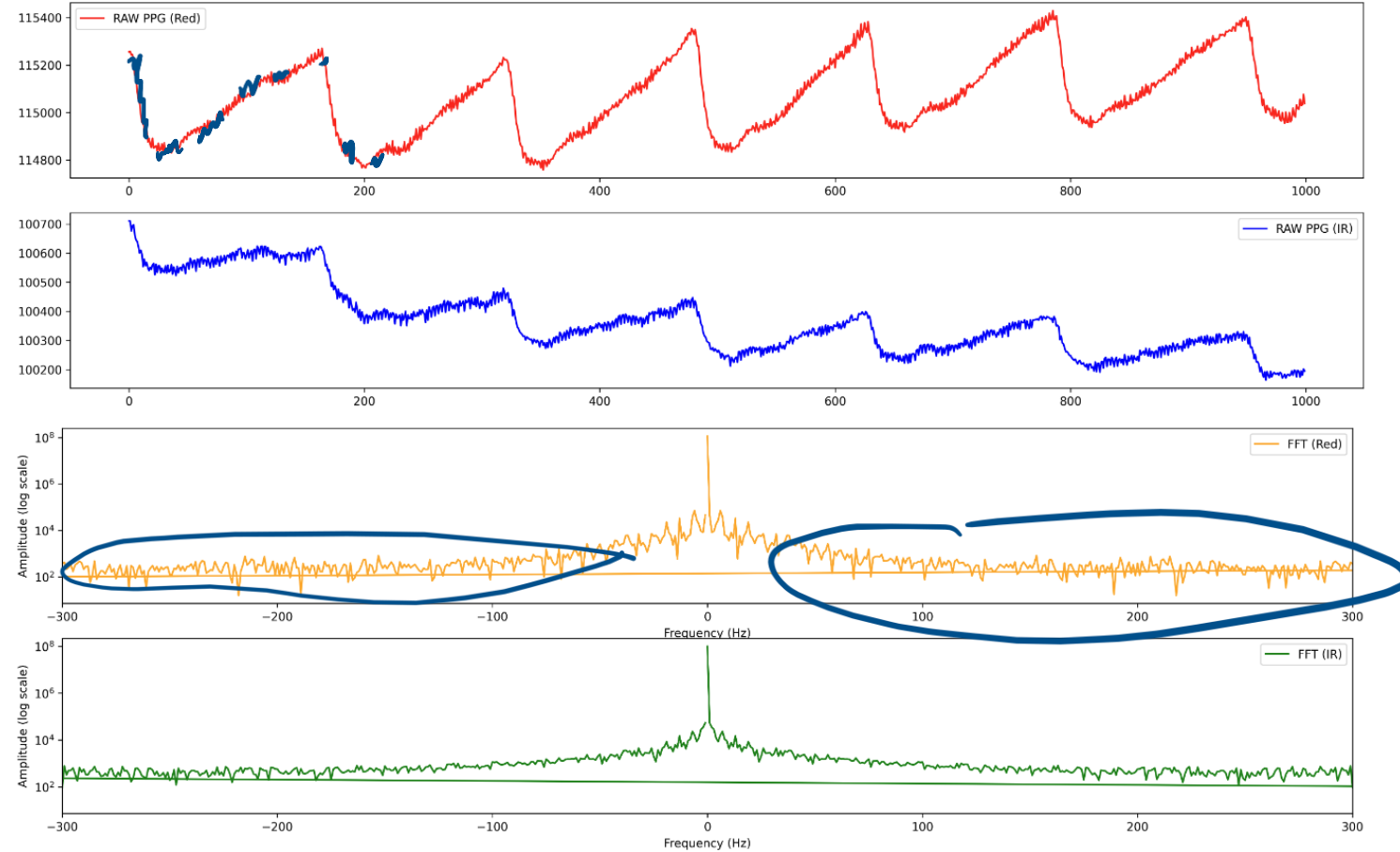
Table 6. SpO₂ Sample Rate Control

SPO2_SR[2:0]	SAMPLES PER SECOND
000	50
001	100
010	200
011	400
100	800
101	1000
110	1600
111	3200

See Table 11 and Table 12 for Pulse Width vs. Sample Rate information.

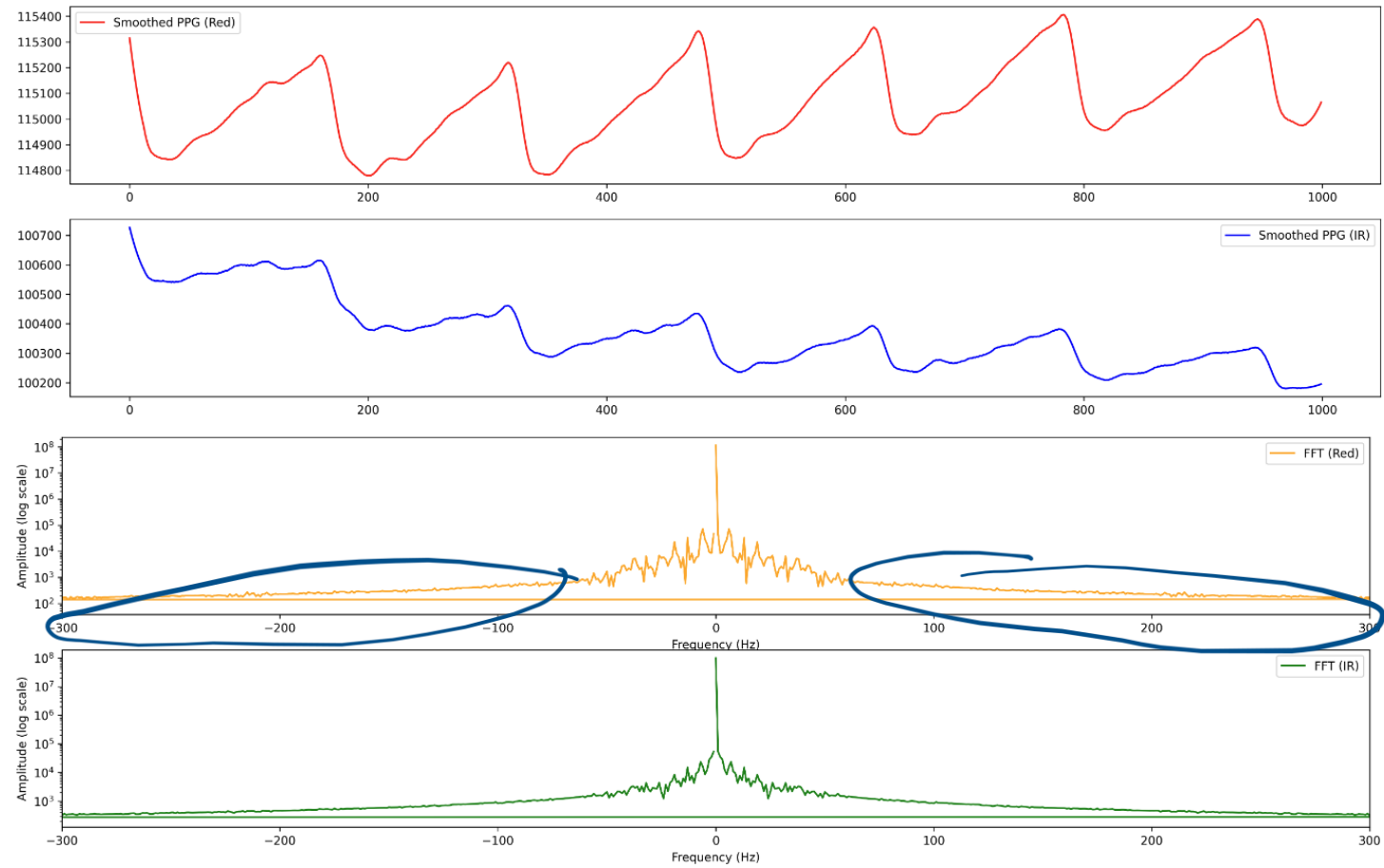
How many
data points

Going back to the PPG data collected

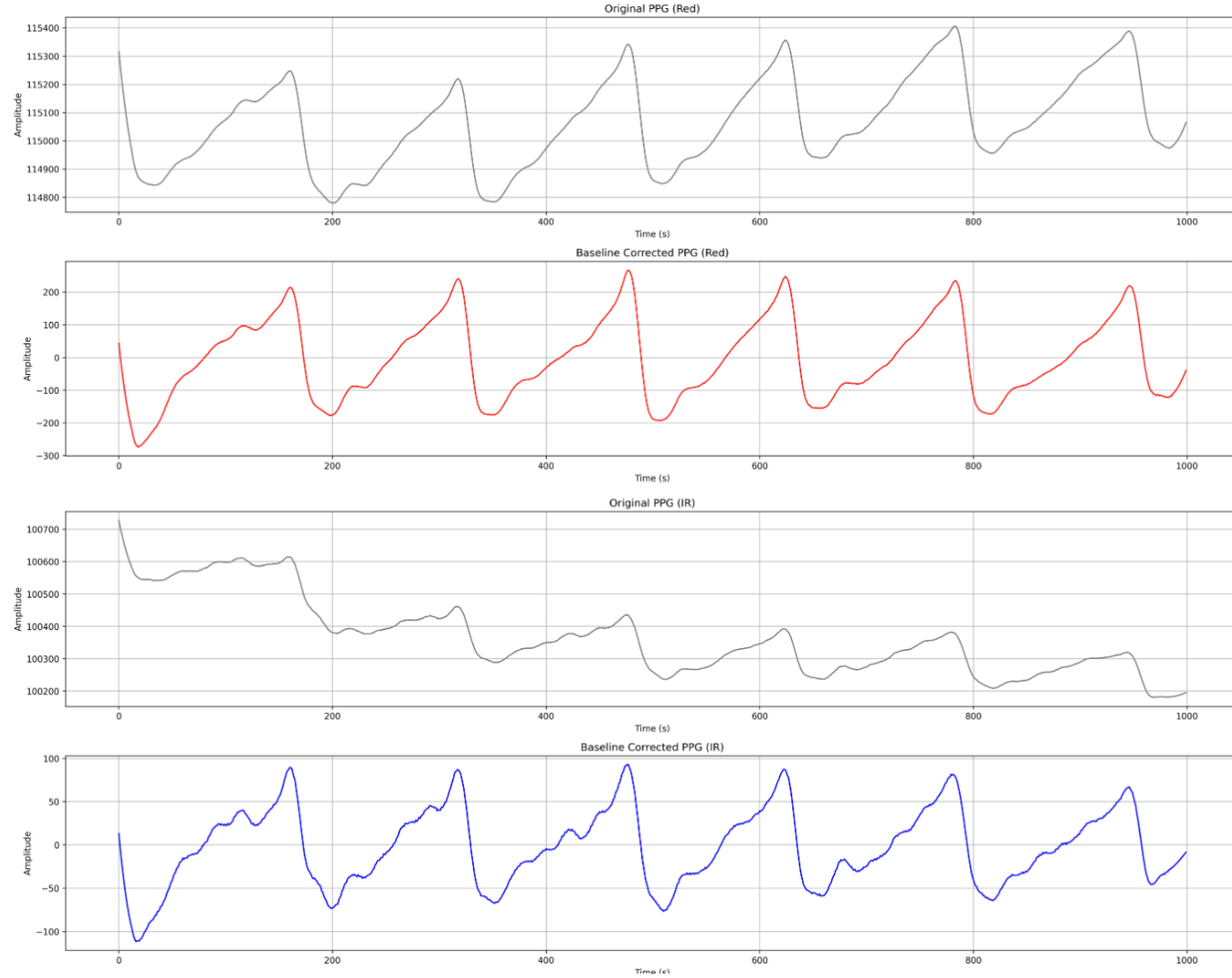


Smoothing

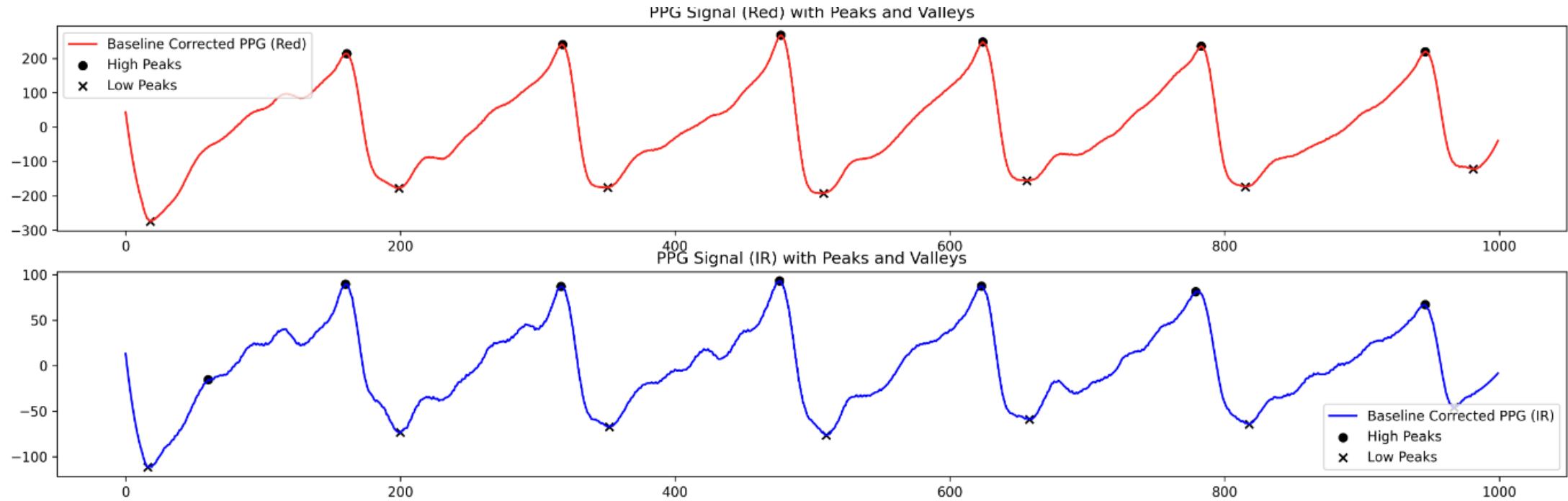
Gets rid of
high freq
noise



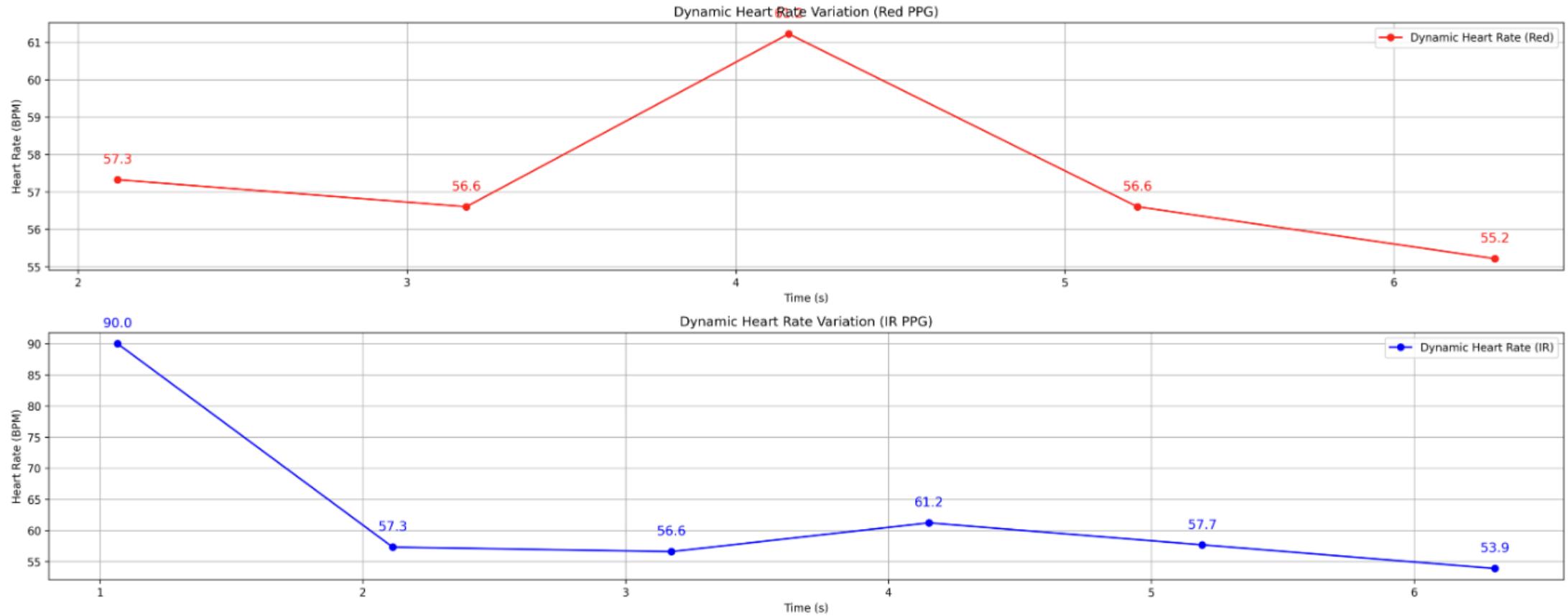
Drift correction



Peak detection



Heart rate



Using single component to find heart rate

